

Chapter 9

Stable and Unstable Manifolds for Equilibria

This chapter is concerned with the dynamics of an ODE near an equilibrium. The fundamental theoretical result, stated below, is the existence of stable and unstable manifolds the nonlinear analogue of the stable and unstable linear subspaces of linear systems discussed in Section 4.3. Our focus is the development of efficient high order approximation methods –with validated error bounds– for these manifolds.

To be more specific, consider a differential equation $\dot{x} = f(x)$ with associated flow $\varphi: \mathbb{R} \times \mathbb{R}^L \rightarrow \mathbb{R}^L$. Assume that $\tilde{x}$ is an equilibrium. The *stable* and *unstable* sets for $\tilde{x}$ are defined to be

$$W^s(\tilde{x}) := \left\{ x \in \mathbb{R}^n \mid \lim_{t \rightarrow \infty} \varphi(t, x) = \tilde{x} \right\} \quad (9.1)$$

$$W^u(\tilde{x}) := \left\{ x \in \mathbb{R}^n \mid \lim_{t \rightarrow -\infty} \varphi(t, x) = \tilde{x} \right\}, \quad (9.2)$$

respectively. A natural question is to characterize the structure of these sets. At this level of generality there is very little that can be said, however a natural first step is to simplify the problem by restricting the dynamics to a neighborhood of $\tilde{x}$.

Definition 9.0.1. Consider a flow $\varphi: \mathbb{R} \times \mathbb{R}^L \rightarrow \mathbb{R}^L$ with an equilibrium point $\tilde{x}$. Given a neighborhood U of $\tilde{x}$ the associated *local* stable and unstable sets are given by

$$W_{\text{loc}}^s(\tilde{x}) \stackrel{\text{def}}{=} W^s(\tilde{x}, U) := \{x \in W^s(\tilde{x}) \mid \varphi([0, \infty), x) \subset U\} \quad (9.3)$$

$$W_{\text{loc}}^u(\tilde{x}) \stackrel{\text{def}}{=} W^u(\tilde{x}, U) := \{x \in W^u(\tilde{x}) \mid \varphi((-\infty, 0], x) \subset U\}, \quad (9.4)$$

respectively.

The following result is fundamental for much of nonlinear dynamical systems.

Theorem 9.0.2. [(Un)Stable Manifold Theorem] *Consider an ordinary differential equation*

$$\dot{x} = f(x), \quad x \in \mathbb{R}^L \quad (9.5)$$

with $f \in C^k(\mathbb{R}^L)$, $k \in \{1, 2, 3, \dots, \infty\} \cup \{\omega\}$ for which $f(0) = 0$. Assume

$$Df(0) = \begin{bmatrix} A^u & 0 \\ 0 & A^s \end{bmatrix}$$

is hyperbolic where $A^u \in M_d(\mathbb{R})$, $A^s \in M_{L-d}(\mathbb{R})$, and the real parts of the all the eigenvalues of A^u and A^s are positive and negative, respectively. Then, there exists a neighborhood $U = U^u \times U^s \subset \mathbb{R}^d \times \mathbb{R}^{L-d}$ of the origin and C^k functions $P^u: U^u \rightarrow \mathbb{R}^L$ and $P^s: U^s \rightarrow \mathbb{R}^L$ that are tangent to $\mathbb{R}^d \times \{0\}$ and $\{0\} \times \mathbb{R}^{L-d}$ at the point 0, respectively, such that

$$W_{\text{loc}}^u(0) = P^u(U^u) \quad \text{and} \quad W_{\text{loc}}^s(0) = P^s(U^s).$$

There are essentially two methods used to prove Theorem 9.0.2: the Lypaunov-Perron method (see [10]), and the Hadamard graph-transform (see [17, 34]). The approach we take, called the Parameterization Method and introduced in Section 9.1, is more restrictive in that it assumes that the vector field f is analytic and that the eigenvalues of the linearized system at the fixed point satisfy certain algebraic constraints. However, this approach allows us to derive an explicit quantitative description of the the local stable and unstable manifolds. We use the Lorenz equations to provide simple examples of the application of the Parameterization Method in Section 9.2, before providing a more general discussion in Section ??.

9.1 The Parameterization Method

Throughout this section we consider the differential equation

$$\dot{x} = f(x), \quad x \in \mathbb{R}^L,$$

under the assumption that $\tilde{x}$ is a fixed point and f is analytic in a neighborhood of $\tilde{x}$. We let $\varphi: \mathbb{R} \times \mathbb{R}^L \rightarrow \mathbb{R}^L$ denote the associated flow. Furthermore, assume that

$$Df(\tilde{x}) = \begin{bmatrix} A^u & 0 \\ 0 & A^s \end{bmatrix}$$

is hyperbolic where $A^u \in M_d(\mathbb{R})$, $A^s \in M_{L-d}(\mathbb{R})$, and the real parts of the all the eigenvalues of A^u and A^s are positive and negative, respectively.

For the purpose of this discussion our goal is to obtain an explicit representation of $W_{\text{loc}}^s(\tilde{x})$, i.e., we want to construct $P: U^s \rightarrow \mathbb{R}^L$ such that $P(U^s) = W_{\text{loc}}^s(\tilde{x})$, where $U^s \subset \mathbb{R}^{L-d}$ is a neighborhood of 0. We will do this by identifying a partial differential equation (PDE) with the property that its solution is the function P .

To motivate the PDE, we denote the flow of the linearized system restricted to the stable linear space

$$\dot{y} = A^s y, \quad y \in \mathbb{R}^{L-d}, \quad (9.6)$$

by $\psi(t, y) = e^{A^s t} y$. Let $B \subset \mathbb{R}^{L-d}$ be a forward invariant neighborhood of the origin (the existence of B follows from Theorem 4.3.3). One way to insure that $P(B) = W_{\text{loc}}^s(\tilde{x})$ is to choose P such that it is a conjugacy between the flow ψ and the flow ϕ where the latter is restricted to $W_{\text{loc}}^s(\tilde{x})$. This choice implies that

$$\varphi(t, P(y)) = P(e^{A^s t} y) \quad \text{for all } y \in B \text{ and } t \geq 0. \quad (9.7)$$

It is easy to check that Observe that this implies that if (9.31) is satisfied, then

$$P(0) = \tilde{x} \quad \text{and} \quad P(B) \subset W_{\text{loc}}^s(\tilde{x}). \quad (9.8)$$

The first equality follows from direct substitute. To understand the inclusion observe that given $y \in B$

$$\lim_{t \rightarrow \infty} \varphi(t, P(y)) = \lim_{t \rightarrow \infty} P(e^{A^s t} y) = P\left(\lim_{t \rightarrow \infty} e^{A^s t} y\right) = P(0) = \tilde{x},$$

since all the eigenvalues of A^s have negative real part. The geometric interpretation of this conjugacy is illustrated in Figure 9.1.

Lemma 9.1.1 (Parameterization Lemma). *A function $P: B \subset \mathbb{R}^{L-d} \rightarrow \mathbb{R}^L$ satisfies the conjugacy relationship (9.31) if and only if P is a solution of the PDE*

$$DP(y)A^s y = f(P(y)), \quad y \in B. \quad (9.9)$$

Proof. ($\Leftarrow$) Suppose that P solves the partial differential equation (9.9) in B . Fix $y \in B$ and $t > 0$. Define the function $\gamma_y: [0, t] \rightarrow \mathbb{R}^L$ by

$$\gamma_y(t) \stackrel{\text{def}}{=} P\left(e^{A^s t} y\right). \quad (9.10)$$

Then, $\gamma_y(0) = P(y)$ and

$$\frac{d}{dt} \gamma_y(t) = \frac{d}{dt} P(e^{A^s t} y) = DP(e^{A^s t} y) A^s e^{A^s t} y = f(P(e^{A^s t} y)) = f(\gamma_y(t)),$$

where we pass from the first to the second equality by the chain rule, from the second to the third equality by the invariance equation (9.9) and the fact that $e^{A^s t} y \in B$ when $t > 0$, and from the third to the fourth equation by the definition of γ_y . Hence γ_y is the solution of the initial value problem

$$\dot{\gamma}_y(t) = f(\gamma_y(t)), \quad \gamma_y(0) = P(y). \quad (9.11)$$

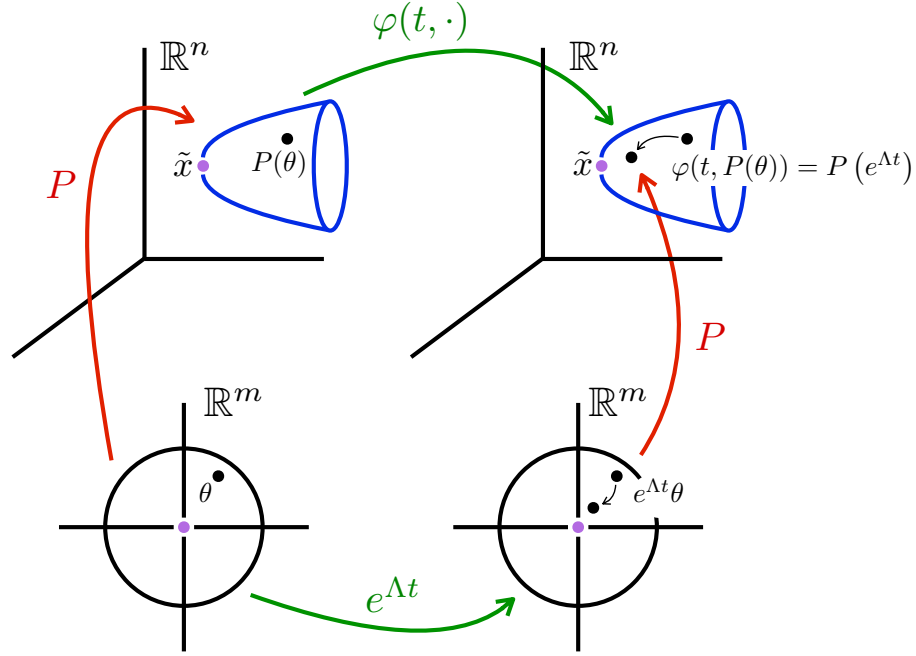


Figure 9.1: **Schematic of the Parameterization Method:** The figure illustrates the conjugacy described by Equation (9.31). The bottom half of the figure represents the parameter space in $\mathbb{R}^m$ (the domain of the parameterization P) while the top half of the figure represents the phase space in $\mathbb{R}^n$. The image of P is the local stable manifold shown in blue. The dynamics are depicted by moving from the left to the right side of the figure. The dynamics in the parameter space is generated by exponentiating the matrix of stable eigenvalues Λ . The dynamics in phase space is generated by the flow φ associated with the vector field f . The diagram *commutes* in the sense that applying first the chart map P and then nonlinear flow $\varphi(t, \cdot)$ is required to be the same as applying the linear dynamics $e^{\Lambda t}$ and then the chart map P . The result is that the dynamics on the local stable manifold are described by the stable linear dynamics.

Therefore by definition $\varphi(t, \gamma_y(0)) = \gamma_y(t)$, and it follows from (9.10) and (9.11) that

$$\varphi(t, P(y)) = P(e^{A^s t} y).$$

($\implies$) Suppose that P satisfies the conjugacy relationship (9.31) for all $y \in B$. Fix $y \in B$ and differentiate both sides with respect to t in order to obtain

$$f(\varphi(t, P(y))) = DP(e^{A^s t} y) A^s e^{A^s t} y.$$

Taking the limit as $t \rightarrow 0$ gives that $P(y)$ is a solution of (9.9). □

In light of Lemma 9.1.1, if we solve the PDE (9.9), then we obtain the desired conjugacy P . In keeping with the theme of this book we will identify a solution to (9.9) via the contraction mapping theorem, which implies that we need to be identifying a unique solution. However, by Theorem 4.3.6 there must be a continuum of solutions to (9.9) as any linear change of variables on $\mathbb{R}^{L-d}$ gives rise to a conjugate flow and hence a different function P . Our perspective is that this provides an opportunity to impose an additional constraint that will lead to a unique solution, but of equal importance allow us to perform the computations in as simple a format as possible.

With this in mind, for the purpose of this introduction, we add an extra assumption. Assume that A^s has real eigenvalues and is diagonalizable. In particular, let $L - d = M$. We denote the eigenvalues of A^s by $\lambda_1, \dots, \lambda_M$ and for $m = 1, \dots, M$ let ξ_m denote a unit eigenvector associated with eigenvalue λ_m . By Proposition 4.2.9 $\{\xi_m \mid m = 1, \dots, M\}$ is a basis for $\mathbb{R}^M$ and in that basis the solution operator to (9.6) takes the form $e^{\Lambda t}$ where

$$\Lambda = \begin{pmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_M \end{pmatrix}. \quad (9.12)$$

Set

$$E = \begin{bmatrix} r_1 \xi^{(1)} & \dots & r_M \xi^{(M)} \end{bmatrix}, \quad (9.13)$$

where $r_m > 0$, i.e., the $M \times M$ matrix with columns given by eigenvectors. We impose the constraint that

$$DP(0) = E, \quad (9.14)$$

which is an explicit means of forcing $W_{\text{loc}}^s(\tilde{x}) = P(B)$ to be tangent to the stable linear subspace associated with $Df(\tilde{x})$.

We remark for future reference that (9.9) can be written more explicitly as

$$\lambda_1 y_1 \frac{\partial}{\partial y_1} P(y_1, \dots, y_M) + \dots + \lambda_M y_M \frac{\partial}{\partial y_M} P(y_1, \dots, y_M) = f(P(y_1, \dots, y_M)) \quad (9.15)$$

for $y = (y_1, \dots, y_M) \in B$.

Remark 9.1.2. The previous discussion focused on deriving (9.9) to identify W_{loc}^s via P . Using the same arguments, but starting with $\dot{y} = A^u y$, one can see that solving (9.9) will produce a function $P: B \rightarrow \mathbb{R}^L$ such that $P(B) = W_{\text{loc}}^u$; the most significant change is that one considers $t \leq 0$ the proof of Lemma 9.1.1.

Of course, we still need to show that we can solve for P , which is the topic of the next section in the context of the Lorenz equations.

9.2 Local Stable and Unstable Manifolds for the Lorenz Equations

Recall from Example 2.4.9 that the Lorenz system

$$\dot{x} = f(x) = \begin{pmatrix} \sigma(x_2 - x_1) \\ \rho x_1 - x_2 - x_1 x_3 \\ -\beta x_3 + x_1 x_2 \end{pmatrix}$$

has three equilibria

$$0 = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \quad \text{and} \quad \tilde{x}^\pm = \begin{pmatrix} \pm \sqrt{\beta(\rho - 1)} \\ \pm \sqrt{\beta(\rho - 1)} \\ \rho - 1 \end{pmatrix}$$

when $\rho > 1$. At the classical parameter values $\sigma = 10$, $\beta = 8/3$ and $\rho = 28$ the fixed point at the origin 0 the eigenvalues are real (one is positive and two are negative) while at the fixed points $\tilde{x}^\pm$ one eigenvalue is negative and two eigenvalues are complex with positive real parts. In the following subsections we demonstrate how Theorem 7.6.2 can be used to compute the one-dimensional unstable manifold $W_{\text{loc}}^u(0)$, the two-dimensional stable manifold $W_{\text{loc}}^s(0)$, and the two-dimensional unstable manifold $W_{\text{loc}}^u(\tilde{x}^\pm)$.

9.2.1 One-dimensional (un)stable manifold for the Lorenz system

Let $\tilde{x}$ be a fixed point for the Lorenz system. Assume $Df(\tilde{x})$ has a unique eigenvalue λ with positive (negative) real part (this implies that λ is real). For future reference let ξ denote a fixed choice of the associated unit eigenvector.

For simplicity (the reader should check that this assumption has no bearing on the discussion that follows) assume $\lambda > 0$. In this case $M = 1$ and P is a function of a single variable that we express through a Taylor series expansion

$$P(y) = \sum_{n=0}^{\infty} a_n y^n = \sum_{n=0}^{\infty} \begin{pmatrix} a_{1,n} \\ a_{2,n} \\ a_{3,n} \end{pmatrix} y^n$$

where $a = \{a_n \in \mathbb{R}^3\}_{n=0}^{\infty} \in (\ell_\nu^1)^3$ for some $\nu > 0$. Keeping in mind our intention to apply Theorem 7.6.2 we set $X = (\ell_\nu^1)^3$.

We now turn to the problem of identifying the function $F: X \rightarrow Y$ for which we want to solve for a zero. By (9.8) $P(0) = \tilde{x}$, hence

$$\begin{pmatrix} a_{1,0} \\ a_{2,0} \\ a_{3,0} \end{pmatrix} = \begin{pmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \tilde{x}_3 \end{pmatrix}.$$

The constraint (9.14) implies that

$$\begin{pmatrix} a_{1,1} \\ a_{2,1} \\ a_{3,1} \end{pmatrix} = \begin{pmatrix} r\xi_1 \\ r\xi_2 \\ r\xi_3 \end{pmatrix}.$$

More generally observe that (9.15) reduces to

$$\frac{\partial P}{\partial y}(y)\lambda y - f(P(y)) = 0$$

or equivalently

$$\lambda \sum_{n=0}^{\infty} \begin{pmatrix} (n+1)a_{1,n+1} \\ (n+1)a_{2,n+1} \\ (n+1)a_{3,n+1} \end{pmatrix} y^{n+1} - \sum_{n=0}^{\infty} \begin{pmatrix} \sigma(a_{2,n} - a_{1,n}) \\ \rho a_{1,n} - a_{2,n} - (a_1 * a_3)_n \\ -\beta a_{3,n} + (a_1 * a_2)_n \end{pmatrix} y^n = 0.$$

Matching like powers suggests that we define $F(a) = (F_1(a), F_2(a), F_3(a))$ by

$$(F_1(a))_n = \begin{cases} a_{1,0}, & \text{if } n = 0, \\ a_{1,1} - r\xi_1, & n = 1, \\ \lambda n a_{1,n} - \sigma(a_{2,n} - a_{1,n}), & n \geq 2 \end{cases} \quad (9.16)$$

$$(F_2(a))_n = \begin{cases} a_{2,0}, & \text{if } n = 0, \\ a_{2,1} - r\xi_2, & n = 1, \\ \lambda n a_{2,n} - \rho a_{1,n} + a_{2,n} + (a_1 * a_3)_n, & n \geq 2 \end{cases} \quad (9.17)$$

$$(F_3(a))_n = \begin{cases} a_{3,0}, & \text{if } n = 0, \\ a_{3,1} - r\xi_3, & n = 1, \\ \lambda n a_{3,n} + \beta a_{3,n} - (a_1 * a_2)_n, & n \geq 2. \end{cases} \quad (9.18)$$

As a consequence of Proposition 10.1.1 we define $Y = (\ell_\omega^1)^3$ where $\omega = \left\{ \frac{\nu^{n+1}}{n+1} \right\}_{n=0}^\infty$.

We have at this point identified X , Y , and $F: X \rightarrow Y$, and thus completed the first step towards the application of Theorem 7.6.2.

The second step is the identification of the approximate solution $\bar{a}$. For any choice of N the value of the first $N+1$ functions $\{(F_l(a))_n \mid n = 0, \dots, N\}$, $l = 1, 2, 3$ are determined entirely by $\{a_{l,n} \mid n = 0, \dots, N, l = 1, 2, 3\}$. Therefore, straightforward truncation defines $F^{(N)}: \mathbb{R}^{3(N+1)} \rightarrow \mathbb{R}^{3(N+1)}$. We claim that we can always solve $F^{(N)}(a) = 0$. To see why, we begin by returning to the Lorenz equations and noting that the linearization at the equilibrium $\tilde{x}$ is given by

$$Df(\tilde{x}) = \begin{bmatrix} -\sigma & \sigma & 0 \\ \rho - \tilde{x}_3 & -1 & -\tilde{x}_1 \\ \tilde{x}_2 & \tilde{x}_1 & -\beta \end{bmatrix}$$

For a fixed $n \geq 2$ we can rewrite (9.16) - (9.18) as

$$\begin{bmatrix} \lambda n + \sigma & -\sigma & 0 \\ -\rho + a_{3,0} & \lambda n + 1 & a_{1,0} \\ -a_{2,0} & -a_{1,0} & \lambda n + \beta \end{bmatrix} \begin{bmatrix} a_{1,n} \\ a_{2,n} \\ a_{3,n} \end{bmatrix} = \begin{bmatrix} 0 \\ \sum_{j=1}^{n-1} a_{1,n-j} a_{3,j} \\ \sum_{j=1}^{n-1} a_{1,n-j} a_{2,j} \end{bmatrix}.$$

Recall that $(a_{1,0}, a_{2,0}, a_{3,0}) = (\tilde{x}_1, \tilde{x}_2, \tilde{x}_3)$, and hence, we can write

$$\left(\begin{bmatrix} \lambda n & 0 & 0 \\ 0 & \lambda n & 0 \\ 0 & 0 & \lambda n \end{bmatrix} + \begin{bmatrix} \sigma & -\sigma & 0 \\ -\rho + \tilde{x}_3 & 1 & \tilde{x}_1 \\ -\tilde{x}_2 & -\tilde{x}_1 & \beta \end{bmatrix} \right) \begin{bmatrix} a_{1,n} \\ a_{2,n} \\ a_{3,n} \end{bmatrix} = \begin{bmatrix} 0 \\ \sum_{j=1}^{n-1} a_{1,n-j} a_{3,j} \\ \sum_{j=1}^{n-1} a_{1,n-j} a_{2,j} \end{bmatrix}$$

$$(Df(\tilde{x}) - n\lambda I) \begin{bmatrix} a_{1,n} \\ a_{2,n} \\ a_{3,n} \end{bmatrix} = \begin{bmatrix} 0 \\ -\sum_{j=1}^{n-1} a_{1,n-j} a_{3,j} \\ -\sum_{j=1}^{n-1} a_{1,n-j} a_{2,j} \end{bmatrix}$$

Recall that we assumed that λ is the unique positive eigenvalue of $Df(\tilde{x})$. Therefore, $n\lambda$ is not an eigenvalue, which implies that $Df(\tilde{x}) - n\lambda I$ is invertible for any $n \geq 2$. Since $(a_{1,1}, a_{2,1}, a_{3,1}) = r(\xi_1, \xi_2, \xi_3)$ are we can inductively solve for $(a_{1,n}, a_{2,n}, a_{3,n})$ beginning with $n = 2$.

Remark 9.2.1. If we are trying to identify an M -dimensional unstable manifold, then $Df(\tilde{x})$ will have M eigenvalues with positive real parts and this simple argument for the invertibility of the matrix of acting on the highest degree terms is not sufficient. We will return to this point in Section 9.4.

Let $\bar{a}^{(N)} = (\bar{a}_0, \bar{a}_1, \dots, \bar{a}_N)$ denote an approximate solution to $F^{(N)}(a) = 0$, which can be derived either via the above mentioned inductive process or via an alternative numerical method. Define

$$\bar{a} := (\bar{a}_0, \bar{a}_1, \dots, \bar{a}_{N-1}, \bar{a}_N, 0, 0, 0, \dots) \in \ell_\nu^1.$$

and observe that

$$\bar{P}(y) := \sum_{n=0}^{\infty} \begin{pmatrix} \bar{a}_{1,n} \\ \bar{a}_{2,n} \\ \bar{a}_{3,n} \end{pmatrix} y^n = \sum_{n=0}^N \begin{pmatrix} \bar{a}_{1,n} \\ \bar{a}_{2,n} \\ \bar{a}_{3,n} \end{pmatrix} y^n$$

is an approximate solution to (9.15).

Remark 9.2.2. By (9.14) r allows us to scale the magnitude of $(a_{1,1}, a_{2,1}, a_{3,1})$, which in turn will impact the values of $a_{l,n}$ for all $n \geq 2$. We return to this point at the end of this subsection.

The third step towards the application of Theorem 7.6.2 is the choice of a linear operator $A^\dagger: X \rightarrow Y$ that approximates $DF(\bar{a})$. Rather than writing the formulas for $DF(\bar{a})$, observe that we can view it as a 3×3 matrix of infinite dimensional matrices, i.e.,

$$DF(a) = \begin{bmatrix} D_1 F_1(a) & D_2 F_1(a) & D_3 F_1(a) \\ D_1 F_2(a) & D_2 F_2(a) & D_3 F_2(a) \\ D_1 F_3(a) & D_2 F_3(a) & D_3 F_3(a) \end{bmatrix}$$

Observe that the diagonal matrices have the form

$$\begin{aligned}
 D_1 F_1(a) &= \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 & \\ 0 & 0 & 2\lambda + \sigma & 0 & \\ 0 & 0 & 0 & 3\lambda + \sigma & \\ \vdots & & & & \ddots \end{bmatrix} \\
 D_2 F_2(a) &= \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 & \\ 0 & 0 & 2\lambda + 1 & 0 & \\ 0 & 0 & 0 & 3\lambda + 1 & \\ \vdots & & & & \ddots \end{bmatrix} \\
 D_3 F_3(a) &= \begin{bmatrix} 1 & 0 & 0 & 0 & \cdots \\ 0 & 1 & 0 & 0 & \\ 0 & 0 & 2\lambda + \beta & 0 & \\ 0 & 0 & 0 & 3\lambda + \beta & \\ \vdots & & & & \ddots \end{bmatrix}
 \end{aligned}$$

where the diagonal terms are increasing at a linear rate. For n large, $n\lambda + \sigma \approx n\lambda$, and thus, more generally, the constant terms σ , 1 , and β are of limited consequence.

The terms of the off-diagonal matrices are bounded, e.g.,

$$\begin{aligned}
 D_2 F_1(a) &= \begin{bmatrix} 0 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 0 & \\ 0 & 0 & -\sigma & 0 & \\ 0 & 0 & 0 & -\sigma & \\ \vdots & & & & \ddots \end{bmatrix}, \\
 D_1 F_2(a) &= \begin{bmatrix} 0 & 0 & 0 & 0 & \cdots \\ 0 & 0 & 0 & 0 & \\ a_{3,2} & a_{3,1} & -\rho + a_{3,0} & 0 & \\ a_{3,3} & a_{3,2} & a_{3,1} & -\rho + a_{3,0} & \\ \vdots & & & & \ddots \end{bmatrix}.
 \end{aligned}$$

Furthermore, $a \in (\ell_\nu^1)^3$ implies that if n is large and k is small relative to n , then $|a_{n,n-k}|$ is negligible. This suggests that for large n the diagonal terms of $D_l F_l(a)$ dominate, and in particular, we can ignore the contribution of $D_k F_l(a)$ for $k \neq l$. Retaining the infinite

matrix notation we choose

$$A^\dagger = \left[\begin{array}{cc|cc|ccc} D_1 F_1^{(N)}(\bar{a}) & \mathbf{0} & D_2 F_1^{(N)}(\bar{a}) & \mathbf{0} & D_3 F_1^{(N)}(\bar{a}) & \mathbf{0} & \dots \\ \mathbf{0} & \bar{\Lambda} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \\ \hline D_1 F_2^{(N)}(\bar{a}) & \mathbf{0} & D_2 F_2^{(N)}(\bar{a}) & \mathbf{0} & D_3 F_2^{(N)}(\bar{a}) & \mathbf{0} & \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \bar{\Lambda} & \mathbf{0} & \mathbf{0} & \\ \hline D_1 F_3^{(N)}(\bar{a}) & \mathbf{0} & D_2 F_3^{(N)}(\bar{a}) & \mathbf{0} & D_3 F_3^{(N)}(\bar{a}) & \mathbf{0} & \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \bar{\Lambda} & \end{array} \right]$$

where $\bar{\Lambda}$ is a diagonal matrix with diagonal entries $(N+k)\lambda$, $k = 1, 2, \dots$

The fourth step towards the application of Theorem 7.6.2 is the choice of a linear operator $A: Y \rightarrow X$ that approximates $(A^\dagger)^{-1}$. Since $F^{(N)}(\bar{a}): \mathbb{R}^{3(N+1)} \rightarrow \mathbb{R}^{3(N+1)}$ is finite dimensional we can numerically identify $A^{(N)}$ an approximation to $(DF^{(N)}(ba))^{-1}$. With the appropriate decomposition of $A^{(N)}$ we can set

$$A = \left[\begin{array}{cc|cc|ccc} A_{1,1}^{(N)} & \mathbf{0} & A_{1,2}^{(N)} & \mathbf{0} & A_{1,3}^{(N)} & \mathbf{0} & \dots \\ \mathbf{0} & \bar{\Lambda}^{-1} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \\ \hline A_{2,1}^{(N)} & \mathbf{0} & A_{2,2}^{(N)} & \mathbf{0} & A_{2,3}^{(N)} & \mathbf{0} & \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \bar{\Lambda}^{-1} & \mathbf{0} & \mathbf{0} & \\ \hline A_{3,1}^{(N)} & \mathbf{0} & A_{3,2}^{(N)} & \mathbf{0} & A_{3,3}^{(N)} & \mathbf{0} & \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \mathbf{0} & \bar{\Lambda}^{-1} & \end{array} \right]$$

To prove that $\bar{a}$ is an approximation of a true solution of (9.9) via Theorem 7.6.2 requires knowledge of the Y_0 , Z_0 , Z_1 and Z_2 bounds. For this it is convenient to have an explicit representation of DF .

$$\frac{\partial F_{1,n}}{\partial a_{1,m}}(a) = \begin{cases} 1 & \text{if } n = 0, 1 \text{ and } n = m \\ 0 & \text{if } n = 0, 1 \text{ and } m \neq n \\ n\lambda + \sigma & \text{if } n = m \text{ and } n \geq 2 \\ 0 & \text{if } m \neq n \text{ and } n \geq 2 \end{cases} \quad (9.19)$$

$$\frac{\partial F_{1,n}}{\partial a_{2,m}}(a) = \begin{cases} 0 & \text{if } n = 0, 1 \\ -\sigma & \text{if } n = m \text{ and } n \geq 2 \\ 0 & \text{if } m \neq n \text{ and } n \geq 2 \end{cases} \quad (9.20)$$

$$\frac{\partial F_{1,n}}{\partial a_{3,m}}(a) = 0 \quad (9.21)$$

$$\frac{\partial F_{2,n}}{\partial a_{1,m}}(a) = \begin{cases} 0 & \text{if } n = 0, 1 \\ a_{3,n-m} & \text{if } n \geq 2 \text{ and } m < n \\ -\rho + a_{3,0} & \text{if } n \geq 2 \text{ and } n = m \\ 0 & \text{if } n \geq 2 \text{ and } m > n \end{cases} \quad (9.22)$$

$$\frac{\partial F_{2,n}}{\partial a_{2,m}}(a) = \begin{cases} 1 & \text{if } n = 0, 1 \text{ and } n = m \\ 0 & \text{if } n = 0, 1 \text{ and } m \neq n \\ n\lambda + 1 & \text{if } n \geq 2 \text{ and } n = m \\ 0 & \text{if } n \geq 2 \text{ and } m \neq n \end{cases} \quad (9.23)$$

$$\frac{\partial F_{2,n}}{\partial a_{3,m}}(a) = \begin{cases} 0 & \text{if } n = 0, 1 \\ a_{1,n-m} & \text{if } n \geq 2 \text{ and } m \leq n \\ 0 & \text{if } n \geq 2 \text{ and } m > n \end{cases} \quad (9.24)$$

$$\frac{\partial F_{3,n}}{\partial a_{1,m}}(a) = \begin{cases} 0 & \text{if } n = 0, 1 \\ -a_{2,n-m} & \text{if } n \geq 2 \text{ and } m \leq n \\ 0 & \text{if } n \geq 2 \text{ and } m > n \end{cases} \quad (9.25)$$

$$\frac{\partial F_{3,n}}{\partial a_{1,m}}(a) = \begin{cases} 0 & \text{if } n = 0, 1 \\ -a_{1,n-m} & \text{if } n \geq 2 \text{ and } m \leq n \\ 0 & \text{if } n \geq 2 \text{ and } m > n \end{cases} \quad (9.26)$$

$$\frac{\partial F_{3,n}}{\partial a_{3,m}}(a) = \begin{cases} 1 & \text{if } n = 0, 1 \text{ and } n = m \\ 0 & \text{if } n = 0, 1 \text{ and } m \neq n \\ n\lambda + \beta & \text{if } n \geq 2 \text{ and } n = m \\ 0 & \text{if } n \geq 2 \text{ and } m \neq n \end{cases} \quad (9.27)$$

Theorem 9.2.3. Fix N and ν . Set

$$\begin{bmatrix} \bar{a}_{1,0} \\ \bar{a}_{2,0} \\ \bar{a}_{3,0} \end{bmatrix} = \begin{bmatrix} \tilde{x}_1 \\ \tilde{x}_2 \\ \tilde{x}_3 \end{bmatrix} \quad \text{and} \quad \begin{bmatrix} \bar{a}_{1,1} \\ \bar{a}_{2,1} \\ \bar{a}_{3,1} \end{bmatrix} = \begin{bmatrix} \xi_1 \\ \xi_2 \\ \xi_3 \end{bmatrix}$$

where ξ is an eigenvector associated with the unique positive eigenvalue λ . Let $F^{(N)}: \mathbb{R}^{3(N+1)} \rightarrow \mathbb{R}^{3(N+1)}$ be defined by (9.16)-(9.18). Let $\bar{a}^{(N)} = (\bar{a}_0, \bar{a}_1, \dots, \bar{a}_N)$ be an approximate solution to $F^{(N)}(a) = 0$ and let $\bar{a} \in (\ell_\nu^1)^3$ be given by

$$(\bar{a}_0, \dots, \bar{a}_N, 0, 0, \dots).$$

Let $A^{(N)}$ be an approximate inverse of $DF^{(N)}(\bar{a})$. Set

$$Y_0 := \max \left\{ Y_0^{(1)}, Y_0^{(2)}, Y_0^{(3)} \right\}$$

where

$$\begin{aligned} Y_0^{(1)} &:= \sum_{n=0}^N \left| \sum_{j=1}^3 \left(A_{1,j}^{(N)} F_j^{(N)}(\bar{a}^{(N)}) \right)_n \right| \nu^n \\ Y_0^{(2)} &:= \sum_{n=0}^N \left| \sum_{j=1}^3 \left(A_{2,j}^{(N)} F_j^{(N)}(\bar{a}^{(N)}) \right)_n \right| \nu^n + \frac{1}{|\lambda|} \sum_{n=N+1}^{2N} \frac{1}{n} \|(\bar{a}_1 * \bar{a}_3)_n\| \nu^n \\ Y_0^{(3)} &:= \sum_{n=0}^N \left| \sum_{j=1}^3 \left(A_{3,j}^{(N)} F_j^{(N)}(\bar{a}^{(N)}) \right)_n \right| \nu^n + \frac{1}{|\lambda|} \sum_{n=N+1}^{2N} \frac{1}{n} \|(\bar{a}_1 * \bar{a}_2)_n\| \nu^n \end{aligned}$$

Set

$$Z_0 \stackrel{\text{def}}{=} \max_{j=1,2,3} \sum_{j'=1}^3 Z_0^{(j,j')}$$

where

$$Z_0^{(j,j')} \stackrel{\text{def}}{=} \max_{0 \leq k' \leq N} \frac{1}{\nu^{k'}} \sum_{0 \leq k \leq N} \left| (B_{j,j'}^{(N)})_{k,k'} \right| \nu^k$$

and

$$B_{j,j'}^{(N)} \stackrel{\text{def}}{=} \delta_{j,j'} I_{\mathbb{R}^{N+1}} - \sum_{n=1}^N A_{j,n}^{(N)} D_{j'} F_n^{(N)}(\bar{a}^{(N)})$$

Finally, set

$$\begin{aligned} Z_1 &:= \frac{1}{|\lambda|(N+1)} \max \{2|\sigma|, |\rho| + 1 + \|\bar{a}_1\|_{1,\nu} + \|\bar{a}_3\|_{1,\nu}, |\beta| + \|\bar{a}_1\|_{1,\nu} + \|\bar{a}_2\|_{1,\nu}\}, \\ Z_2 &:= 2\|A\|_{B((\ell_\nu^1)^3)} \end{aligned}$$

Then, Y_0 , Z_0 , Z_1 and Z_2 satisfy the hypothesis of Theorem 7.6.2.

As these bounds are a special case of the general bounds presented in Section 9.4 we do not provide a proof of Theorem 9.2.3, but use them to present some specific examples. Consider the classical parameter values $\sigma = 10$, $\beta = 8/3$ and $\rho = 28$. The origin has a one-dimensional unstable manifold while $(\pm\sqrt{\beta(\rho-1)}, \pm\sqrt{\beta(\rho-1)}, \rho-1)$ have a one-dimensional stable manifolds. Fix $N = 50$, and set $r = 40$. For each of the three equilibria, we used Newton's method to obtain $\bar{a}^{(N)} \in \mathbb{R}^{3(N+1)} = \mathbb{R}^{153}$ such that $\|F^{(N)}(\bar{a}^{(N)})\|_1 \leq \text{tol} = 8 \times 10^{-13}$. The one-dimensional manifolds are plotted in Figure 9.2.

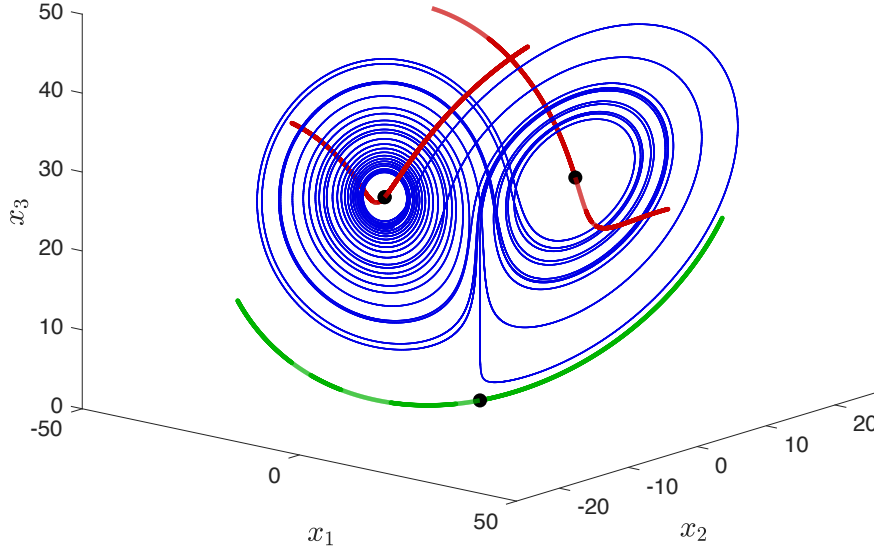


Figure 9.2: One-dimensional (un)stable manifolds in the Lorenz equations. In green, the unstable manifold of the origin. In red, the one-dimensional stable manifolds of the eyes. In blue, the *strange attractor* in the Lorenz system. Each numerical approximation is obtained by fixing the Taylor order to $N = 50$ and by fixing the length of the eigenvector to 40.

9.2.2 Two-dimensional stable manifold for the origin in Lorenz

Continuing to use the Lorenz equations as a model we now approximate $W_{\text{loc}}^s(0)$, which is a two-dimensional manifold as the eigenvalues of $DF(0)$ are

$$\lambda_1 = -\beta < 0, \quad \lambda_2 = \frac{-(\sigma + 1) - \sqrt{(\sigma + 1)^2 + 4\sigma(\rho - 1)}}{2} < 0,$$

and

$$\lambda_3 = \frac{-(\sigma + 1) + \sqrt{(\sigma + 1)^2 + 4\sigma(\rho - 1)}}{2} > 0.$$

We use the parameterization method, i.e., we express the stable manifold via a function $P: B \subset \mathbb{R}^2 \rightarrow \mathbb{R}^3$ of the form

$$P(y) = \sum_{|\alpha| \geq 0} a_\alpha y^\alpha$$

where $a_\alpha \in \mathbb{R}^3$ and $\alpha = (\alpha_1, \alpha_2)$ is a multi-index. Since the stable manifold is two-dimensional, (9.15) takes the form

$$\lambda_1 y_1 \frac{\partial}{\partial y_1} P(y_1, y_2) + \lambda_2 y_2 \frac{\partial}{\partial y_2} P(y_1, y_2) = f(P(y_1, y_2))$$

or equivalently

$$\sum_{|\alpha|=1}^{\infty} (\alpha_1 \lambda_1 + \alpha_2 \lambda_2) a_{\alpha} y^{\alpha} = \sum_{|\alpha|=0}^{\infty} \begin{pmatrix} \sigma(a_{2,\alpha} - a_{1,\alpha}) \\ \rho a_{1,\alpha} - a_{2,\alpha} - (a_1 * a_3)_{\alpha} \\ -\beta a_{3,\alpha} + (a_1 * a_2)_{\alpha} \end{pmatrix} y^{\alpha} = \sum_{|\alpha|=0}^{\infty} \phi_{\alpha}(a) y^{\alpha}. \quad (9.28)$$

By (9.8) $P(0) = \tilde{x}$, hence the $|\alpha| = 0$ term takes the values

$$\begin{pmatrix} a_{1,\alpha} \\ a_{2,\alpha} \\ a_{3,\alpha} \end{pmatrix} = \begin{pmatrix} a_{1,0,0} \\ a_{2,0,0} \\ a_{3,0,0} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

The constraint (9.14) implies that

$$\begin{pmatrix} a_{1,1,0} & a_{1,0,1} \\ a_{2,1,0} & a_{2,0,1} \\ a_{3,1,0} & a_{3,0,1} \end{pmatrix} = \begin{pmatrix} r_1 \xi_1^{(1)} & r_2 \xi_1^{(2)} \\ r_1 \xi_2^{(1)} & r_2 \xi_2^{(2)} \\ r_1 \xi_3^{(1)} & r_2 \xi_3^{(2)} \end{pmatrix}.$$

where $\xi^{(1)}$ and $\xi^{(2)}$ are a unit eigenvalues for λ_1 and λ_2 , respectively. Therefore, a_{α} is fixed for $|\alpha| = 0, 1$.

Preparing for the application of Theorem 7.6.2 we observe that matching powers in (9.28) leads to the infinite set of equations

$$(\alpha_1 \lambda_1 + \alpha_2 \lambda_2) a_{\alpha} = \phi_{\alpha}(a), \quad |\alpha| = \alpha_1 + \alpha_2 \geq 2.$$

This is equivalent to

$$\begin{pmatrix} \alpha_1 \lambda_1 + \alpha_2 \lambda_2 + \sigma & \sigma & 0 \\ a_{3,0,0} & \alpha_1 \lambda_1 + \alpha_2 \lambda_2 + 1 & a_{1,0,0} \\ a_{2,0,0} & a_{1,0,0} & \alpha_1 \lambda_1 + \alpha_2 \lambda_2 + \beta \end{pmatrix} \begin{pmatrix} a_{1,\alpha} \\ a_{2,\alpha} \\ a_{3,\alpha} \end{pmatrix} = \psi_{\alpha}(a)$$

$$\begin{pmatrix} \alpha_1 \lambda_1 + \alpha_2 \lambda_2 + \sigma & \sigma & 0 \\ 0 & \alpha_1 \lambda_1 + \alpha_2 \lambda_2 + 1 & 0 \\ 0 & 0 & \alpha_1 \lambda_1 + \alpha_2 \lambda_2 + \beta \end{pmatrix} \begin{pmatrix} a_{1,\alpha} \\ a_{2,\alpha} \\ a_{3,\alpha} \end{pmatrix} = \psi_{\alpha}(a)$$

where $\psi_{\alpha}(a)$ is composed of terms $a_{\alpha'}$ where $|\alpha'| < |\alpha|$. Thus, if

$$\begin{pmatrix} \alpha_1 \lambda_1 + \alpha_2 \lambda_2 + \sigma & \sigma & 0 \\ 0 & \alpha_1 \lambda_1 + \alpha_2 \lambda_2 + 1 & 0 \\ 0 & 0 & \alpha_1 \lambda_1 + \alpha_2 \lambda_2 + \beta \end{pmatrix} \begin{pmatrix} a_{1,\alpha} \\ a_{2,\alpha} \\ a_{3,\alpha} \end{pmatrix}$$

is invertible, then we can solve for a inductively. For this system this is equivalent to the question of whether

$$-\beta \alpha_1 + \frac{-(\sigma + 1) - \sqrt{(\sigma + 1)^2 + 4\sigma(\rho - 1)}}{2} \alpha_2 \in \{\sigma, 1, \beta\}$$

for some non-negative integers α_1 and α_2 . For the classical parameter values the classical parameter values $\sigma = 10$, $\beta = 8/3$ and $\rho = 28$, this cannot happen so we can solve for P .

To explain this idea in the simplest possible setting assume that the stable eigenvalues $\{\lambda_1, \dots, \lambda_m\}$ are real and that to each eigenvalue λ_i there is an associated eigenvector ξ_i . Then we can construct the following matrices: an $m \times m$ diagonal matrix with the diagonal entries made up of the stable eigenvalues

$$\Lambda = \begin{pmatrix} \lambda_1 & \dots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \dots & \lambda_m \end{pmatrix} \quad (9.29)$$

and an $n \times m$ matrix whose columns are the associated eigenvectors

$$A_0 = [\xi_1 | \dots | \xi_m]. \quad (9.30)$$

Using the basis defined by the stable eigenvectors, the linearized equation for (9.5) restricted to the stable subspace takes the form

$$\dot{y} = \Lambda y, \quad y \in \mathbb{R}^m.$$

The associated flow is given by $\psi(t, x) = e^{\Lambda t} x$. As indicated earlier our goal is to construct an analytic function $P: B^m \rightarrow \mathbb{R}^n$ such that $P(B^m) = W_{\text{loc}}^s(\tilde{x})$, where $B^m \subset \mathbb{R}^m$ is a neighborhood of 0. To obtain constraints, so that we can solve for P , we begin by insisting that P be a conjugacy between the flow φ of (9.5) restricted to $W_{\text{loc}}^s(\tilde{x})$ and the flow ψ of the linear equation. The most obvious restriction is that P must map fixed points to fixed points and hence

$$P(0) = \tilde{x}.$$

To obtain the conjugacy we assume that

$$DP(0) = A_0$$

and

$$\varphi(t, P(\theta)) = P(e^{\Lambda t} \theta), \quad (9.31)$$

for all $\theta \in B^m$. The geometric meaning of this conjugacy is illustrated in Figure 9.1. To see that $P(B^m) \subset W_{\text{loc}}^s(\tilde{x})$ observe that

$$\lim_{t \rightarrow \infty} \varphi(t, P(\theta)) = \lim_{t \rightarrow \infty} P(e^{\Lambda t} \theta) = P\left(\lim_{t \rightarrow \infty} e^{\Lambda t} \theta\right) = P(0) = \tilde{x},$$

since the entries of Λ are negative.

Note that any function $P(\theta)$ satisfying Equation (9.31) is one-to-one on B^m . To see this observe that P is tangent to the stable eigenspace at the origin as $DP(0) = A_0$. Moreover recall that A_0 is of full rank as its columns are linearly independent. By the

implicit function theorem P is of rank m , and hence one-to-one, in some neighborhood $U \subset B^m$ of 0. Now suppose that $\theta_1, \theta_2 \in B^m$ and that $P(\theta_1) = P(\theta_2)$. Then for any $t \in \mathbb{R}$, $\varphi(t, P(\theta_1)) = \varphi(t, P(\theta_2))$ by the uniqueness of the initial value problem. Choose $T > 0$ large enough so that $e^{\Lambda T}\theta_1, e^{\Lambda T}\theta_2 \in U$. By the conjugacy relation we have that $P(e^{\Lambda T}\theta_1) = P(e^{\Lambda T}\theta_2)$, and since the arguments are in U , the local immersion gives that $e^{\Lambda T}\theta_1 = e^{\Lambda T}\theta_2$. But $e^{\Lambda T}$ is an isomorphism and we have $\theta_1 = \theta_2$.

The utility of (9.31) is limited by the appearance of the flow φ in the equation. In practice the flow is only known implicitly, that is it is determined by solving the differential equation. The following lemma establishes a more practical infinitesimal version of (9.31).

Equation (9.9) can be written more explicitly as

$$\lambda_1 \theta_1 \frac{\partial}{\partial \theta_1} P(\theta_1, \dots, \theta_m) + \dots + \lambda_m \theta_m \frac{\partial}{\partial \theta_m} P(\theta_1, \dots, \theta_m) = f(P(\theta_1, \dots, \theta_m))$$

for $\theta = (\theta_1, \dots, \theta_m) \in B^m$.

Remark 9.2.4. Let $g : \mathbb{R}^m \rightarrow \mathbb{R}^m$ be a vector field and $P : \mathbb{R}^m \rightarrow \mathbb{R}^n$ with $m < n$ be a smooth one-to-one map. The image of P is a smooth manifold which we denote by $\mathcal{M}$, and there is a vector field defined on $\mathcal{M}$ by $DP(\theta)g(\theta)$. This is called *the push forward of g by P* .

The geometric interpretation of (9.9) is that the push forward of the linear vector field $\Lambda\theta$ by P is equal to the vector field on the image of P . Then, intuitively speaking, we expect that P “lifts” orbits in B^m of the vector field $\Lambda\theta$ to orbits on $P(B^m)$ of the vector field f .

9.2.3 Two-dimensional stable manifold for the origin in Lorenz

Let us return now to the Lorenz system in a case we ignored above: namely, when we have two eigenvalues of like stability. More specifically, we focus on the two-dimensional stable manifold of $\tilde{x} = 0 \in \mathbb{R}^3$. Note that the stable eigenvalues of $Df(0)$ are given by

$$\lambda_1 = -\beta < 0, \quad \lambda_2 = \frac{-(\sigma+1) - \sqrt{(\sigma+1)^2 + 4\sigma(\rho-1)}}{2} < 0$$

while the unstable eigenvalue is given by $\lambda_3 = \frac{-(\sigma+1) + \sqrt{(\sigma+1)^2 + 4\sigma(\rho-1)}}{2} > 0$.

Denote by $\xi_1 = (0, 0, 1)^T$ and ξ_2 the two associated stable eigenvectors. Recalling (9.30), denote by $A_0 = (\xi_1, \xi_2)$ the (3×2) -dimensional matrix.

With $m = 2$ the invariance equation (9.9) is

$$\lambda_1 \theta_1 \frac{\partial}{\partial \theta_1} P(\theta_1, \theta_2) + \lambda_2 \theta_2 \frac{\partial}{\partial \theta_2} P(\theta_1, \theta_2) = f[P(\theta_1, \theta_2)], \quad (9.32)$$

while the first order constraints (??) yield that

$$a_{0,0} = 0 \in \mathbb{R}^3, \quad a_{1,0} = \xi_1 \quad \text{and} \quad a_{0,1} = \xi_2. \quad (9.33)$$

Since P is a function of two variables, we will look for a power series solution in two variables. Analytic functions of several complex variables are discussed in more detail below. For now, proceeding formally we assume that

$$P(\theta) = P(\theta_1, \theta_2) = \sum_{|\alpha| \geq 0} a_\alpha \theta^\alpha = \sum_{\alpha_1=0}^{\infty} \sum_{\alpha_2=0}^{\infty} \begin{pmatrix} (a_1)_{\alpha_1, \alpha_2} \\ (a_2)_{\alpha_1, \alpha_2} \\ (a_3)_{\alpha_1, \alpha_2} \end{pmatrix} \theta_1^{\alpha_1} \theta_2^{\alpha_2}$$

and try again to work out the unknown power series coefficients. Here, we use the multi-indices notation, that is, given $\alpha = (\alpha_1, \alpha_2) \in \mathbb{N}^2$ and $\theta = (\theta_1, \theta_2) \in \mathbb{R}^2$, $|\alpha| = \alpha_1 + \alpha_2$ and $\theta^\alpha = \theta_1^{\alpha_1} \theta_2^{\alpha_2}$.

First, we note that

$$\lambda_1 \theta_1 \frac{\partial}{\partial s_1} P(\theta_1, \theta_2) + \lambda_2 \theta_2 \frac{\partial}{\partial \theta_2} P(\theta_1, \theta_2) = \sum_{|\alpha|=0}^{\infty} (\alpha_1 \lambda_1 + \alpha_2 \lambda_2) a_\alpha \theta^\alpha$$

and that for the Lorenz vector field

$$f(P(\theta)) = \sum_{|\alpha|=0}^{\infty} \phi_\alpha(a) \theta^\alpha = \sum_{|\alpha|=0}^{\infty} \begin{pmatrix} (\phi_1(a))_\alpha \\ (\phi_2(a))_\alpha \\ (\phi_3(a))_\alpha \end{pmatrix} \theta^\alpha \stackrel{\text{def}}{=} \sum_{|\alpha|=0}^{\infty} \begin{pmatrix} \sigma((a_2)_\alpha - (a_1)_\alpha) \\ \rho(a_1)_\alpha - (a_2)_\alpha - (a_1 * a_3)_\alpha \\ -\beta(a_3)_\alpha + (a_1 * a_2)_\alpha \end{pmatrix} \theta^\alpha,$$

where $a_1 * a_2$ and $a_1 * a_3$ denote two-dimensional Cauchy products. Denote $a \stackrel{\text{def}}{=} (a_1, a_2, a_3)$.

Matching like powers leads to

$$(\alpha_1 \lambda_1 + \alpha_2 \lambda_2) a_\alpha = \phi_\alpha(a),$$

for $|\alpha| = \alpha_1 + \alpha_2 \geq 2$.

Combining the above derivations lead to introducing the zero finding problem $F(a) = 0$, where each component of $F(a) = (F_1(a), F_2(a), F_3(a))$ is given component-wise by

$$(F_j(a))_\alpha = \begin{cases} (a_j)_{0,0} - \tilde{x}_j, & \alpha = (0, 0), \\ (a_j)_{1,0} - (\xi_1)_j, & \alpha = (1, 0), \\ (a_j)_{0,1} - (\xi_2)_j, & \alpha = (0, 1), \\ (\alpha_1 \lambda_1 + \alpha_2 \lambda_2)(a_j)_\alpha - (\phi_j(a))_\alpha, & |\alpha| \geq 2. \end{cases} \quad (9.34)$$

Given a sequence $a_j = ((a_j)_\alpha)_{|\alpha| \geq 0}$ and fixing a Taylor order N , consider the finite dimensional projection of a_j to be $a_j^{(N)} = ((a_j)_\alpha)_{|\alpha| \leq N}^N \in \mathbb{R}^{(N+1)(N+2)/2}$. Projecting each a_j and F_j for $j = 1, 2, 3$, this leads to the finite dimensional reduction $F^{(N)} : \mathbb{R}^{3(N+1)(N+2)/2} \rightarrow \mathbb{R}^{3(N+1)(N+2)/2}$. Applying Newton's method, we may obtain an approximate zeros $\bar{a}^{(N)} =$

$(\bar{a}_1^{(N)}, \bar{a}_2^{(N)}, \bar{a}_2^{(N)})$ such that $F^{(N)}(\bar{a}^{(N)}) \approx 0$. This yields a numerical approximation of the two-dimensional stable manifold given by the truncated Taylor series

$$P^{(N)}(s) \stackrel{\text{def}}{=} \sum_{|\alpha|=0}^N \bar{a}_\alpha s^\alpha = \sum_{|\alpha|=0=0}^N \begin{pmatrix} (\bar{a}_1)_\alpha \\ (\bar{a}_2)_\alpha \\ (\bar{a}_3)_\alpha \end{pmatrix} s^\alpha. \quad (9.35)$$

We fixed the classical parameter values $\sigma = 10$, $\beta = 8/3$ and $\rho = 28$, fixed the Taylor order to be $N = 20$, and fixed the length of ξ_1 to be 12 and the length of ξ_2 to be 14. We used Newton's method to compute $\bar{a}^{(N)} \in \mathbb{R}^{3(N+1)(N+2)/2} = \mathbb{R}^{693}$ such that $\|F^{(N)}(\bar{a}^{(N)})\|_1 \leq \text{tol} = 4 \times 10^{-12}$. We plot the two-dimensional manifold in Figure 9.3.

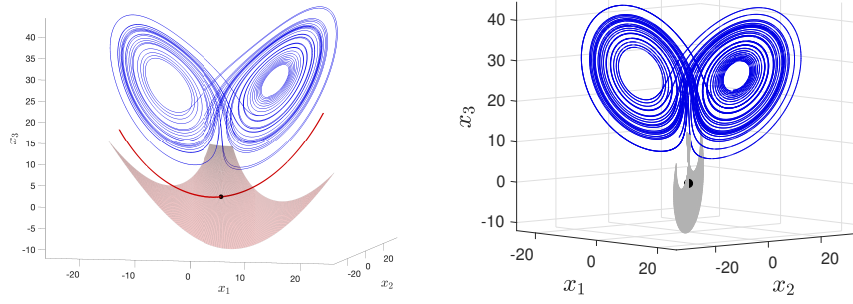


Figure 9.3: Two different views of the two-dimensional stable manifold of the origin in the Lorenz equations obtained with a Taylor approximation of order $N = 20$. In red, the one-dimensional unstable manifolds of the origin, as computed in the previous section. In blue, the *strange attractor* in the Lorenz system.

9.2.4 The case of complex eigenvalues in the Lorenz equations

Recall that the eyes $\tilde{x}^\pm \stackrel{\text{def}}{=} (\pm\sqrt{\beta(\rho-1)}, \pm\sqrt{\beta(\rho-1)}, \rho-1)$ have two-dimensional unstable manifolds. Let us choose the positive eye $\tilde{x}^+$ as an equilibrium solution, and denote by $\lambda_1, \lambda_2 \in \sigma(Df(\tilde{x}^+))$ the two unstable complex conjugate eigenvalues, that is with $\bar{\lambda}_1 = \lambda_2$. Note that this case has not been treated at the beginning of Section 9.1, where we assumed that the eigenvalues were real. However, we now demonstrate how the parameterization method can be adapted to this case.

Denote the two complex conjugate eigenvectors ξ_1 and ξ_2 , associated respectively to λ_1, λ_2 with $\bar{\xi}_1 = \xi_2$.

We now look for a parameterization of the form $P : B^2 \subset \mathbb{C}^2 \rightarrow \mathbb{C}^3$ such that the invariance equation (9.32) holds, that is

$$\lambda_1 \theta_1 \frac{\partial}{\partial \theta_1} P(\theta) + \lambda_2 \theta_2 \frac{\partial}{\partial \theta_2} P(\theta) = f(P(\theta)),$$

where $\theta = (\theta_1, \theta_2) \in \mathbb{C}^2$, $\lambda_1, \lambda_2 = \overline{\lambda_1} \in \mathbb{C}$ and while the first order constraints (??) are now given by

$$a_{0,0} = \tilde{x}^+ \in \mathbb{R}^3, \quad a_{1,0} = \xi_1 \in \mathbb{C}^3 \quad \text{and} \quad a_{0,1} = \xi_2 = \overline{\xi_1} \in \mathbb{C}^3.$$

The idea now is to consider the expansion

$$P(\theta) = \sum_{|\alpha|=0}^{\infty} a_{\alpha} \theta^{\alpha}, \quad \theta \in B^2 \subset \mathbb{C}^2 \quad (9.36)$$

The remaining coefficients a_{α} for $|\alpha| \geq 2$ are worked out by plugging the unknown power series (9.36) into the invariance equation (9.9), matching like powers and isolating the highest order terms in order to obtain recurrence relations for the α -th coefficient. It can be shown that this formal computation results in the linear system of equations

$$[Df(\tilde{x}^+) - (\alpha_1 \lambda_1 + \alpha_2 \lambda_2)I] a_{\alpha} = R_{\alpha}, \quad (9.37)$$

which is known as the *homological equation* for the parameterization coefficients. Here R_{α} is a function of only multi-indices β with $|\beta| < |\alpha|$. The specific form of R_{α} is determined by the nonlinearity of f . Now one can check that, due to the symmetry properties of the homological equation (9.37), the power series coefficients of P satisfy

$$a_{(\alpha_2, \alpha_1)} = \overline{a_{\alpha}} = \overline{a_{(\alpha_1, \alpha_2)}},$$

where the bar denotes complex conjugate. In other words for every complex coefficient which appears in the power series expansion of P , its complex conjugate appears as well.

Using a similar zero finding problem as the one in (9.34), we may apply the radii polynomial approach to obtain the parameterization as given in (9.36). Once we have the parameterization $P: B^2 \rightarrow \mathbb{C}^3$ of the complex stable manifold we define the map Q in a real domain $B^2 \subset \mathbb{R}^2$ by making the complex conjugate change of variables

$$Q(\psi_1, \psi_2) = P(\psi_1 + i\psi_2, \psi_1 - i\psi_2) = \sum_{|\alpha|=0}^{\infty} a_{\alpha} (\psi_1 + i\psi_2)^{\alpha_1} (\psi_1 - i\psi_2)^{\alpha_2}. \quad (9.38)$$

Then evaluating P on complex conjugate variables results in a real value and the image of Q is real, since

$$\begin{aligned} \overline{Q(\psi_1, \psi_2)} &= \overline{\sum_{|\alpha|=0}^{\infty} a_{\alpha} (\psi_1 + i\psi_2)^{\alpha_1} (\psi_1 - i\psi_2)^{\alpha_2}} = \sum_{|\alpha|=0}^{\infty} \overline{a_{\alpha}} (\psi_1 - i\psi_2)^{\alpha_1} (\psi_1 + i\psi_2)^{\alpha_2} \\ &= \sum_{|\alpha|=0}^{\infty} a_{(\alpha_2, \alpha_1)} (\psi_1 - i\psi_2)^{\alpha_1} (\psi_1 + i\psi_2)^{\alpha_2} = \sum_{|\alpha|=0}^{\infty} a_{(\alpha_2, \alpha_1)} (\psi_1 + i\psi_2)^{\alpha_2} (\psi_1 - i\psi_2)^{\alpha_1} \\ &= \sum_{|\alpha|=0}^{\infty} a_{(\alpha_1, \alpha_2)} (\psi_1 + i\psi_2)^{\alpha_1} (\psi_1 - i\psi_2)^{\alpha_2} = Q(\psi_1, \psi_2), \end{aligned}$$

where the penultimate equality follows simply by flipping α_1 and α_2 in the sum, resulting in the same value.

We fixed the classical parameter values $\sigma = 10$, $\beta = 8/3$ and $\rho = 28$, fixed the Taylor order to be $N = 20$, and fixed both length of ξ_1 and ξ_2 to be 2. We used Newton's method to compute $\bar{a}^{(N)} \in \mathbb{R}^{3(N+1)(N+2)/2} = \mathbb{R}^{693}$ such that $\|F^{(N)}(\bar{a}^{(N)})\|_1 \leq \text{tol} = 8 \times 10^{-13}$. We plot the two-dimensional manifold in Figure 9.4.

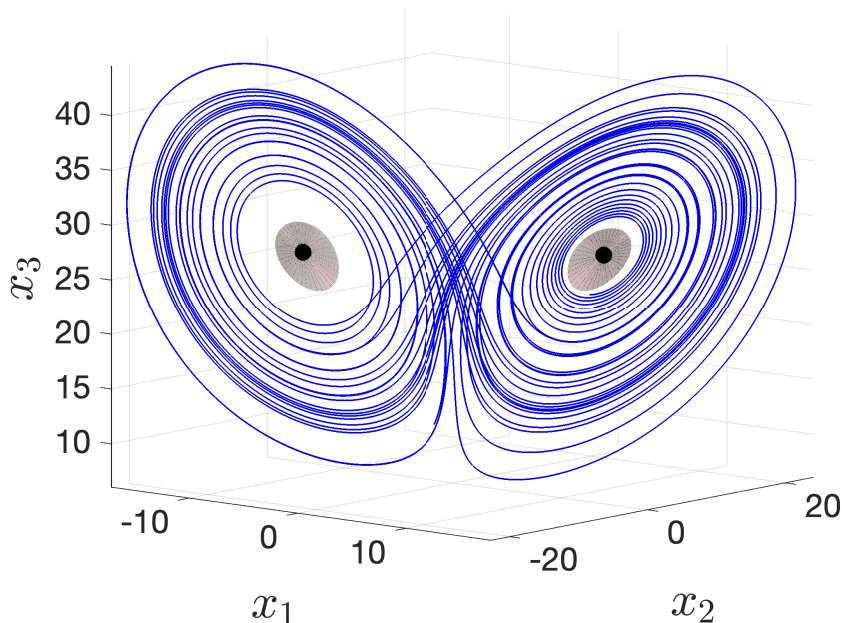


Figure 9.4: Two-dimensional unstable manifolds of the eyes in the Lorenz equations obtained with a Taylor approximation of order $N = 20$. In blue, the *strange attractor* in the Lorenz system.

9.3 Banach algebra for multivariable analytic functions

As discussed in the previous section we use the parameterization method to obtain explicit approximations of the stable and unstable manifolds of hyperbolic fixed points. However, to make use of this we need to show that the formal power series obtained by solving the zero finding problem converges and we need to obtain estimates of the errors obtained from truncations.

Since the stable and unstable manifolds are multidimensional we need to work with multidimensional functions. The following notation is used to simplify expressions in-

volving multidimensional variables.. A m -dimensional multi-index has the form $\alpha = (\alpha_1, \dots, \alpha_m) \in \mathbb{N}^m$. The order or norm of α is given by $|\alpha| := \alpha_1 + \dots + \alpha_m$ and $\alpha! := \alpha_1! \dots \alpha_m!$.

Let $z = (z_1, \dots, z_m) \in \mathbb{C}^m$. We endow $\mathbb{C}^m$ with the norm

$$\|z\| = \max_{1 \leq j \leq m} |z_j|,$$

where $|z_j|$ is the usual complex absolute value. We are interested in approximating functions $g: \mathbb{C}^m \rightarrow \mathbb{C}$ via polynomial expressions. With this in mind, given $z \in \mathbb{C}^m$ and a multi-index $\alpha = (\alpha_1, \dots, \alpha_m)$ define z raised to the α power by

$$z^\alpha := z_1^{\alpha_1} z_2^{\alpha_2} \dots z_m^{\alpha_m}.$$

Let $\Omega \subset \mathbb{C}^m$. A function $g: \Omega \rightarrow \mathbb{C}$ is *analytic* if and only if for each $\hat{z} \in \Omega$ there is an $r > 0$ so that $B_r(\hat{z}) \subset \Omega$ and for all $z \in B_r(\hat{z})$

$$g(z) = \sum_{\alpha_1=0}^{\infty} \dots \sum_{\alpha_m=0}^{\infty} a_{\alpha_1, \dots, \alpha_m} (z_1 - \hat{z}_1)^{\alpha_1} \dots (z_m - \hat{z}_m)^{\alpha_m},$$

where $a_{\alpha_1, \dots, \alpha_m} \in \mathbb{C}$ and the series converges absolutely and uniformly for all $z \in \overline{B_r(\hat{z})}$. The set of all analytic functions $g: \Omega \rightarrow \mathbb{C}$ is denoted by $C^\omega(\Omega)$.

Using multi-index notation this expression can be reduced to

$$g(z) = \sum_{|\alpha|=0}^{\infty} a_\alpha (z - z_0)^\alpha.$$

As in the one dimensional case the *Taylor coefficients* of an analytic function are determined by Cauchy integrals. More precisely, suppose that $g: \Omega \rightarrow \mathbb{C}$ is analytic and $\overline{B_r(\hat{z})} \subset \Omega$. The α -th Taylor coefficient of the power series expansion of g at $\hat{z}$ is given by the iterated line integral

$$a_\alpha := \frac{1}{(2\pi i)^m} \int_{|z_1 - \hat{z}_1|=r} \dots \int_{|z_m - \hat{z}_m|=r} \frac{g(z_1, \dots, z_m)}{(z_1 - \hat{z}_1)^{\alpha_1+1} \dots (z_m - \hat{z}_m)^{\alpha_m+1}} dz_1 \dots dz_m,$$

where the curves $|z_j - \hat{z}_j| = r$, $1 \leq j \leq m$ are parameterized with with positive orientation. Just as in the one dimensional case, the function g is uniquely determined by its Taylor coefficients and in particular if Ω is connected then these coefficients are all zero if and only if g is identically zero on Ω .

Using multi-index notation to describe partial differentiation the α -th partial derivative of g at $\hat{z}$ takes the form

$$\frac{\partial^{|\alpha|}}{\partial z^\alpha} g(z_0) := \frac{\partial^{\alpha_1 + \dots + \alpha_m}}{\partial z_1^{\alpha_1} \dots \partial z_m^{\alpha_m}} g(z_0).$$

If g is analytic on Ω then so is the function $\partial^{|\alpha|}/\partial z^\alpha g$ for every $\alpha \in \mathbb{N}^m$. Moreover, the power series of the partial derivatives are given by “term-by-term” differentiation of the power series for g and we have the expression

$$a_\alpha = \frac{1}{\alpha!} \frac{\partial^{|\alpha|}}{\partial z^\alpha} g(\hat{z}),$$

for the Taylor series coefficients.

Lemma 9.3.1 (Cauchy Estimates). *Let $\Omega \subset \mathbb{C}^m$ and assume that $g: \Omega \rightarrow \mathbb{C}$ is analytic. If $\overline{B_r(\hat{z})} \subset \Omega$, then*

$$|a_\alpha| \leq \frac{1}{r^{|\alpha|}} \sup_{z \in \overline{B_r(\hat{z})}} |g(z)|.$$

As in the one-dimensional case we want to have a Taylor transform that maps us from analytic functions to a Banach algebra.

Let $\mathcal{S}_m$ denote the set of all m -dimensional infinite multi-sequences of complex numbers,

$$\mathcal{S}_m := \{a = \{a_\alpha\} \in \mathbb{C} \mid \alpha = (\alpha_1, \dots, \alpha_m) \in \mathbb{N}^m\}.$$

Note that $\mathcal{S}_m$ is a vector space. For $\nu > 0$ define

$$\|a\|_{1,\nu} := \sum_{|\alpha|=0}^{\infty} |a_\alpha| \nu^{|\alpha|} = \sum_{\alpha_1=0}^{\infty} \dots \sum_{\alpha_m=0}^{\infty} |a_{\alpha_1, \dots, \alpha_m}| \nu^{\alpha_1 + \dots + \alpha_m}.$$

Proposition 9.3.2. *The set*

$$\ell_\nu^1 := \{a \in \mathcal{S}_m \mid \|a\|_{1,\nu} < \infty\} \tag{9.39}$$

is a Banach algebra with norm $\|\cdot\|_{1,\nu}$ and product $$: $\ell_\nu^1 \times \ell_\nu^1 \rightarrow \ell_\nu^1$ given by*

$$(a * b)_\alpha = \sum_{\beta_1=0}^{\alpha_1} \dots \sum_{\beta_m=0}^{\alpha_m} a_{\alpha_1-\beta_1, \dots, \alpha_m-\beta_m} b_{\beta_1, \dots, \beta_m},$$

We refer to $*$ as the *Cauchy product*.

Remark 9.3.3. If $g_1, g_2 \in C^\omega(B_r(\hat{z}))$ have Taylor series expansions

$$g_1(z) = \sum_{|\alpha|=0}^{\infty} (a_1)_\alpha z^\alpha \quad \text{and} \quad g_2(z) = \sum_{|\alpha|=0}^{\infty} (a_2)_\alpha z^\alpha,$$

then $g_1 \cdot g_2 \in C^\omega(B_r(\hat{z}))$ and has power series representation

$$(g_1 \cdot g_2)(z) = \sum_{|\alpha|=0}^{\infty} (a_1 * a_2)_\alpha z^\alpha,$$

i.e. point-wise multiplication in the function domain corresponds to Cauchy products in sequence space.

Theorem 9.3.4. Let $a = \{a_\alpha\}_{\alpha \in \mathbb{N}^m} \in \ell_\nu^1$. If g is defined by

$$g(z) = \sum_{|\alpha|=0}^{\infty} a_\alpha z^\alpha,$$

then $g \in C^\omega(\overline{B_r(\hat{z})})$.

Theorem 9.3.5. If $g \in C^\omega(\overline{B_r(\hat{z})})$, then the Taylor coefficients of g are in ℓ_ν^1 for all $\nu < r$.

Lemma 9.3.6 (Cauchy Bounds in Several Complex Variables). Let $\nu > 0$, and $a = \{a_\alpha\}_{\alpha \in \mathbb{N}^m} \in \ell_\nu^1$. Define $g: B_\nu(0) \rightarrow \mathbb{C}$ by

$$g(z) := \sum_{|\alpha|=0}^{\infty} a_\alpha z^\alpha.$$

Let $0 < \sigma \leq 1$ and

$$\tilde{\nu} = e^{-\sigma} \nu.$$

Then

$$\sup_{z \in B_{\tilde{\nu}}(0)} \left| \frac{\partial}{\partial z_j} g(z) \right| \leq \frac{1}{\nu \sigma} \|a\|_{1,\nu},$$

for each $1 \leq j \leq m$.

9.4 Stable/unstable manifolds for general vector fields

In this section we generalize the Taylor series approach for the Lorenz equations presented in Sections 9.2.1, 9.2.3 and 9.2.4 to general polynomial vector fields in $\mathbb{R}^n$. Consider the vector field

$$\dot{x} = f(x), \tag{9.40}$$

where $f = (f_1, \dots, f_n)$ is a polynomial vector field. For definiteness, we denote by M the maximum degree of the polynomials $f_1, \dots, f_n$, and write

$$f(x) = \sum_{|\alpha|=0}^M c_\alpha x^\alpha \quad \text{with } c_\alpha \in \mathbb{R}^n,$$

where $\alpha \in \mathbb{N}^n$ is a multi-index, $|\alpha| = \alpha_1 + \dots + \alpha_n$, and $x^\alpha = x_1^{\alpha_1} \dots x_n^{\alpha_n}$.

We begin by making assumptions.

- A1. Assume that $\tilde{x} \in \mathbb{R}^n$ is a hyperbolic equilibrium of (9.40).
- A2. Assume that the Jacobian matrix $Df(\tilde{x})$ is diagonalizable over the complex, that is there is a basis of eigenvectors spanning $\mathbb{C}^n$.

Denote by $\lambda_1, \dots, \lambda_m$ the *stable eigenvalues* of $Df(\tilde{x})$ (i.e. $\operatorname{Re}(\lambda_j) < 0$) with $\xi_1, \dots, \xi_m$ some associated *stable eigenvectors*.

From now on, we focus on the computation of a local stable manifold, which we denote by $W_{\text{loc}}^s(\tilde{x})$, and note that $\dim W_{\text{loc}}^s(\tilde{x}) = m \leq n$. The computation of the unstable manifold is similar. The idea of the computational approach is to represent the chart of the local stable manifold using a Taylor series representation $P: B^m \subset \mathbb{C}^m \rightarrow \mathbb{C}^n$ of the form

$$P(\theta) = \sum_{|\alpha|=0}^{\infty} a_{\alpha} \theta^{\alpha}, \quad a_{\alpha} \in \mathbb{C}^n, \quad (9.41)$$

where $B^m \subset \mathbb{C}^m$ is a domain (usually chosen to be a ball) on which the Taylor series converges, and where $\alpha = (\alpha_1, \dots, \alpha_m) \in \mathbb{N}^m$, $|\alpha| = \alpha_1 + \dots + \alpha_m$, $\theta = (\theta_1, \dots, \theta_m) \in \mathbb{R}^m$ and $\theta^{\alpha} = \theta_1^{\alpha_1} \dots \theta_m^{\alpha_m}$.

From the Parameterization Lemma (Lemma 9.1.1), we look for solutions of the PDE

$$DP(\theta)\Lambda\theta = f(P(\theta)), \quad P(0) = \tilde{x}, \quad DP(0) = A_0, \quad (9.42)$$

in the form of a power series (9.41), where $\Lambda \stackrel{\text{def}}{=} \operatorname{diag}(\lambda_1, \dots, \lambda_m)$ and $A_0 = (\xi_1, \dots, \xi_m)$ is an $n \times m$ complex valued matrix. Looking for the solution of (9.42) in the form of power series requires making an extra assumption, which involves the notion of resonances.

Definition 9.4.1. The eigenvalues $\lambda_1, \dots, \lambda_m$ are said to have a *resonance* of order $\alpha = (\alpha_1, \dots, \alpha_m) \in \mathbb{N}^m$ if

$$\alpha_1 \lambda_1 + \dots + \alpha_m \lambda_m - \lambda_j = 0, \quad (9.43)$$

for some $j \in \{1, \dots, m\}$ with $|\alpha| \geq 2$. If there are no resonances at any order $|\alpha| \geq 2$, then the eigenvalues $\lambda_1, \dots, \lambda_m$ are said to be *non-resonant*.

We are ready to state our third hypothesis.

A3. Assume that the stable eigenvalues $\lambda_1, \dots, \lambda_m$ are *non-resonant*

It is shown in [6] that if the vector field f is analytic then the invariance equation (9.42) has an analytic solution P as long as the eigenvalues are non-resonant. Moreover the power series coefficients of P are uniquely determined up to the choice of the scalings of the eigenvectors $\xi_1, \dots, \xi_m$. This abstract result does not however provide explicit bounds on the size of the domain of analyticity B_{ν} for the parameterization: hence the need for a-posteriori validation of our numerical computations. We also note that if there is a resonance then the invariance equation can be modified so that we conjugate to a polynomial (instead of linear) dynamical system [6, 39], and that the later work just cited implements computer assisted error bounds for the resonant case using the radii polynomial approach.

Under the assumption that the hypotheses A1-A3 are satisfied, our goal is to obtain solutions of the form $P(\theta) = (P_1(\theta), \dots, P_n(\theta))$ with power series expansion given in (9.41).

For each $j = 1, \dots, n$, define $a_j \stackrel{\text{def}}{=} \{(a_j)_\alpha\}_{|\alpha| \geq 0}$ and $a \stackrel{\text{def}}{=} (a_1, \dots, a_n)$. If f is a polynomial vector field, then the j -th component of $f(P(\theta))$ is of the form

$$f_j(P(\theta)) = \sum_{|\alpha| \geq 0} (\phi_j(a))_\alpha \theta^\alpha, \quad (9.44)$$

where the α -th Taylor coefficient $(\phi_j(a))_\alpha$ of $f_j(P(\theta))$ is a combination of Cauchy products involving the terms of a . For each $j = 1, \dots, n$, define $\phi_j(a) \stackrel{\text{def}}{=} \{(\phi_j(a))_\alpha\}_{|\alpha| \geq 0}$ and $\phi(a) \stackrel{\text{def}}{=} (\phi_1(a), \dots, \phi_n(a))$.

Using similar ideas that the ones presented in Chapter 8, $\phi(a)$ can be written in terms of the general expression (9.40) for $f(x)$ as follows:

$$\phi(a) = \sum_{|\alpha|=0}^M c_\alpha a_1^{\alpha_1} * \dots * a_n^{\alpha_n} \quad \text{with } c_\alpha \in \mathbb{R}^n, \quad (9.45)$$

which is just the polynomial f with multiplications re-interpreted as Cauchy products. We will from now on simply write

$$\phi(a) = \sum_{|\alpha|=0}^M c_\alpha a^\alpha, \quad a^\alpha \stackrel{\text{def}}{=} a_1^{\alpha_1} * \dots * a_n^{\alpha_n}. \quad (9.46)$$

Recalling (9.42), for each $j = 1, \dots, n$, we need to solve

$$\begin{aligned} DP_j(\theta)\Lambda\theta - f_j(P(\theta)) &= \sum_{|\alpha| \geq 0} (\alpha_1 \lambda_1 + \dots + \alpha_m \lambda_m)(a_j)_\alpha \theta^\alpha - \sum_{|\alpha| \geq 0} (\phi_j(a))_\alpha \theta^\alpha \\ &= \sum_{|\alpha| \geq 0} ((\alpha \cdot \lambda)(a_j)_\alpha - (\phi_j(a))_\alpha) \theta^\alpha \\ &= 0, \end{aligned}$$

where $\alpha \cdot \lambda \stackrel{\text{def}}{=} \alpha_1 \lambda_1 + \dots + \alpha_m \lambda_m$.

For each $j = 1, \dots, n$, let

$$F_j(a) = \{F_j(a)_\alpha\}_{|\alpha| \geq 0}$$

be defined component-wise by

$$(F_j(a))_\alpha \stackrel{\text{def}}{=} \begin{cases} (a_j)_0 - \tilde{x}_j & \text{for } |\alpha| = 0, \\ (a_j)_{e_i} - (\xi_i)_j & \text{for } |\alpha| = 1 \text{ with } \alpha = e_i \text{ for } i = 1, \dots, m, \\ (\alpha \cdot \lambda)(a_j)_\alpha - (\phi_j(a))_\alpha & \text{if } |\alpha| \geq 2, \end{cases} \quad (9.47)$$

where e_i is the i^{th} vector of the canonical basis of $\mathbb{R}^m$.

Recall the definition of ℓ_ν^1 in (9.39). Having introduced a zero finding problem $F = 0$, the main goal is now to show that there exists $\tilde{a} \in X \stackrel{\text{def}}{=} (\ell_\nu^1)^n$ such that $F(\tilde{a}) = 0$. This requires first introducing some notation to deal efficiently with finite truncations and tails, and introducing some basic functional analytic background.

9.4.1 Projections, truncations and functional analytic background

We first consider a single copy of ℓ_ν^1 for any $\nu > 0$. For any $N \in \mathbb{N}$ we define the projection

$$(\pi_N^{1,\nu} a)_\alpha \stackrel{\text{def}}{=} \begin{cases} a_\alpha & \text{for } 0 \leq |\alpha| \leq N, \\ 0 & \text{for } |\alpha| > N, \end{cases} \quad \text{for any } a \in \ell_\nu^1.$$

The projection on the (natural) complement is denoted by

$$\pi_\infty^{1,\nu} a \stackrel{\text{def}}{=} a - \pi_N^{1,\nu} a.$$

We extend these projections to

$$(\ell_\nu^1)^n = \{a = (a_1, \dots, a_n) \mid a_j \in \ell_\nu^1 \text{ for } j = 1, \dots, n\}$$

by setting

$$\pi_N a \stackrel{\text{def}}{=} (\pi_N^{1,\nu} a_1, \dots, \pi_N^{1,\nu} a_n) \quad \text{for any } a \in (\ell_\nu^1)^n,$$

and $\pi_\infty = I_{(\ell_\nu^1)^n} - \pi_N$. To reduce notational burden we choose not to include the dimension n in the notation for these projections. It should always be clear from the context which n is meant. The ranges of the projections π_N and π_∞ are

$$\begin{aligned} X_N^{n,\nu} &= \{a \in (\ell_\nu^1)^n \mid (a_j)_\alpha = 0 \text{ for any } |\alpha| > N \text{ and all } j = 1, \dots, n\}, \\ X_\infty^{n,\nu} &= \{a \in (\ell_\nu^1)^n \mid (a_j)_\alpha = 0 \text{ for all } 0 \leq |\alpha| \leq N \text{ and all } j = 1, \dots, n\}. \end{aligned}$$

Denote

$$N_1 = N_1(N, m) \stackrel{\text{def}}{=} \frac{(N+1)(N+2) \cdots (N+m)}{m!} = \#\{\alpha \in \mathbb{N}^m \mid 0 \leq |\alpha| \leq N\}. \quad (9.48)$$

The subspace $X_N^{1,\nu}$ can be identified with $\mathbb{R}^{N_1}$. Similarly, the subspace $X_N^{n,\nu}$ can be identified with $(\mathbb{R}^{N_1})^n \simeq \mathbb{R}^{nN_1}$. To be explicit, the bijection $\iota_N^{n,\nu} : X_N^{n,\nu} \rightarrow (\mathbb{R}^{N_1})^n$ is given by

$$\iota_N^{n,\nu} a \stackrel{\text{def}}{=} \{(a_j)_\alpha \mid 1 \leq j \leq n, 1 \leq |\alpha| \leq N\}.$$

If there is no ambiguity, then we will drop the superscripts and denote $X_N = X_N^{n,\nu}$, $X_\infty = X_\infty^{n,\nu}$ and $\iota_N = \iota_N^{n,\nu}$; its inverse is denoted by ι_N^{-1} .

We endow the space $X = (\ell_\nu^1)^n$ with the norm

$$\|a\|_X = \max_{1 \leq j \leq n} \|a_j\|_{1,\nu}. \quad (9.49)$$

The projections onto the components of $a = (a_1, \dots, a_n) \in (\ell_\nu^1)^n$ are denoted by

$$\pi^j a = a_j \quad \text{for } j = 1, \dots, n.$$

Let us now briefly discuss linear functionals and operators on ℓ_ν^1 . In the $m = 1$ case we could think of linear operators as infinite matrices, which is very intuitive. In case the dimension of the manifold is $m \geq 2$ this is not possible. To make sense out of linear operators on ℓ_ν^1 we should proceed a little more cautiously.

Suppose that $c = \{c_\alpha\}_{\alpha \in \mathbb{N}^m} \in \mathcal{S}_m$ and define

$$\|c\|_{\infty, \nu^{-1}} = \sup_{|\alpha| \geq 0} \frac{|c_\alpha|}{\nu^{|\alpha|}} \stackrel{\text{def}}{=} \sup_{\alpha \in \mathbb{N}^m} \frac{|c_{\alpha_1, \dots, \alpha_m}|}{\nu^{\alpha_1 + \dots + \alpha_m}}.$$

Let

$$\ell_{\nu^{-1}}^\infty := \{c \in \mathcal{S}_m : \|c\|_{\infty, \nu^{-1}} < \infty\}.$$

This defines another Banach space.

Note that if $a \in \ell_\nu^1$ and $c \in \ell_{\nu^{-1}}^\infty$ then the sum

$$l_c(a) := \sum_{|\alpha|=0}^{\infty} a_\alpha c_\alpha, \tag{9.50}$$

converges. To see this we just note that

$$|l_c(a)| = \left| \sum_{|\alpha|=0}^{\infty} a_\alpha c_\alpha \right| \leq \sum_{|\alpha|=0}^{\infty} |a_\alpha| |c_\alpha| \frac{\nu^{|\alpha|}}{\nu^{|\alpha|}} \leq \sup_{|\alpha| \geq 0} \frac{|c_\alpha|}{\nu^{|\alpha|}} \sum_{|\alpha|=0}^{\infty} |a_\alpha| \nu^{|\alpha|} \leq \|c\|_{\infty, \nu^{-1}} \|a\|_{1, \nu} < \infty.$$

In fact, when we think of $a \in \ell_\nu^1$ as a variable we see that $l_c(a)$ defines a linear map from ℓ_ν^1 into $\mathbb{C}$, i.e. l_c is a linear functional on ℓ_ν^1 . Taking the supremum over all $a \in \ell_\nu^1$ with $\|a\|_{1, \nu} = 1$ shows that l_c is a bounded linear functional. In other words, $l_c \in (\ell_\nu^1)^*$ is in the Banach dual of ℓ_ν^1 .

Observe that Lemma 7.3.11 implies that if $c \in \ell_{\nu^{-1}}^\infty$, then $l_c \in (\ell_\nu^1)^*$, dual space of ℓ_ν^1 . The deeper fact is that $\ell_{\nu^{-1}}^\infty$ is the dual of ℓ_ν^1 , as we describe in the following theorem, whose proof is similar to the proof of Theorem 7.3.10.

Theorem 9.4.2. $\ell_{\nu^{-1}}^\infty$ is isometrically isomorphic to $(\ell_\nu^1)^*$.

For Banach spaces of infinite sequences, the dual space tells us how to understand more general linear operators from the space to itself. So, suppose that $A: \ell_\nu^1 \rightarrow \ell_\nu^1$ is a bounded linear map. Then A assigns to any $a \in \ell_\nu^1$ a new sequence $A(a) = \{A(a)_\alpha\}_{\alpha \in \mathbb{N}^m} \in \ell_\nu^1$. But then we think of each component $A(a)_\alpha$ as a bounded linear map from ℓ_ν^1 into $\mathbb{C}$: in other words each $A(a)_\alpha$ is in the dual space, and has representation

$$A(a)_\alpha = \sum_{|\beta|=0}^{\infty} a_\beta c_\beta^\alpha,$$

for some sequence $c^\alpha \in \ell_{\nu^{-1}}^\infty$. In other words, while we do not think of A as an infinite matrix, we can think of it as having “entries” $c^\alpha = \{c_\beta^\alpha\}_{\beta \in \mathbb{N}^m}$ for all $\alpha \in \mathbb{N}^m$. Or to put it another way, an operator $A: \ell_\nu^1 \rightarrow \ell_\nu^1$ is defined by a sequence of numbers

$$\{c_\beta^\alpha\}_{\alpha, \beta \in \mathbb{N}^m}.$$

So if $a \in \ell_\nu^1$ is a sequence indexed by an m -dimensional multi-index, then a linear operator is described by a sequence indexed by twice as many indices.

This helps us to compute the norm of a linear operator in a convenient way.

Theorem 9.4.3. *Suppose that $A: \ell_\nu^1 \rightarrow \ell_\nu^1$ is a bounded linear operator. Then*

$$\|A\|_{B(\ell_\nu^1)} \leq \sum_{|\alpha|=0}^{\infty} \sup_{|\beta| \geq 0} \frac{|c_\beta^\alpha|}{\nu^{|\beta|}} \nu^{|\alpha|}.$$

Proof. Suppose that $\{c_\beta^\alpha\}_{\alpha, \beta \in \mathbb{N}^m}$ are the coefficients of A . Then for each fixed $\alpha \in \mathbb{N}^m$ the sequence $c^\alpha = \{c_\beta^\alpha\}_{\beta \in \mathbb{N}^m} \in \ell_{\nu^{-1}}^\infty$. Consider

$$\begin{aligned} \|A\|_{B(\ell_\nu^1)} &= \sup_{\|a\|_{1, \nu}=1} \|A(a)\|_{1, \nu} \\ &= \sup_{\|a\|_{1, \nu}=1} \sum_{|\alpha|=0}^{\infty} |A(a)_\alpha| \nu^{|\alpha|} \\ &= \sup_{\|a\|_{1, \nu}=1} \sum_{|\alpha|=0}^{\infty} \left| \sum_{|\beta|=0}^{\infty} a_\beta c_\beta^\alpha \right| \nu^{|\alpha|} \\ &\leq \sup_{\|a\|_{1, \nu}=1} \sum_{|\alpha|=0}^{\infty} \|c^\alpha\|_{\infty, \nu^{-1}} \|a\|_{1, \nu} \nu^{|\alpha|} \\ &= \sum_{|\alpha|=0}^{\infty} \sup_{|\beta| \geq 0} \frac{|c_\beta^\alpha|}{\nu^{|\beta|}} \nu^{|\alpha|}. \end{aligned}$$

□

9.4.2 Fixed point formulation

Having defined in (9.47) a zero finding problem whose solution provide the Taylor coefficients of the power series parameterization $P: B^m \subset \mathbb{C}^m \rightarrow \mathbb{C}^n$ given in (9.41), we turn to the computer-assisted proof of existence of solutions of $F = 0$. This is done once again by using the radii polynomial approach. This first requires assuming that we have an approximate solution $\bar{a} \in X_N$, which necessitates considering a finite dimensional projection.

Recalling the definition of $N_1 = N_1(N, m)$ in (9.48), we define the finite dimensional truncation $F^{(N)}: \mathbb{R}^{nN_1} \rightarrow \mathbb{R}^{nN_1}$ of the full map F by

$$F^{(N)}(a^{(N)}) = \iota_N \pi_N F(\iota_N^{-1} a^{(N)}) \quad \text{for } a^{(N)} \in \mathbb{R}^{nN_1}.$$

In practice we find $\bar{a}$ by finding numerically an approximate zero $\bar{a}^{(N)}$ of $F^{(N)}$. We then denote by $\bar{a} = \iota_N^{-1} \bar{a}^{(N)}$ the approximate zero of F .

Using the identification of X_N and $\mathbb{R}^{n_{N1}}$ via ι_N , as well as the (finite dimensional) Jacobian $DF^{(N)}$, we define the operator $A_N^\dagger : X_N \rightarrow X_N$ by

$$A_N^\dagger := \iota_N^{-1} DF^{(N)}(\bar{a}^{(N)}) \iota_N,$$

and it follows that

$$A_N^\dagger = \pi_N DF(\bar{a}) \quad \text{for any } \bar{a} \in X_N.$$

We also define the operator $A_\infty^\dagger : X_\infty \rightarrow X_\infty$ by

$$(\pi^j A_\infty^\dagger b)_\alpha = (\alpha \cdot \lambda)(\pi^j b)_\alpha.$$

We now set

$$A^\dagger b = A_N^\dagger \pi_N b + A_\infty^\dagger \pi_\infty b.$$

Next we turn our attention to defining an approximate inverse A of $A^\dagger$. With $A^{(N)}$ a numerically computed approximate inverse of the matrix $DF^{(N)}(\bar{a}^{(N)})$, we set $A_N : X_N \rightarrow X_N$ to be the approximate inverse of $A_N^\dagger$ defined by

$$A_N = \iota_N^{-1} A^{(N)} \iota_N.$$

We define $A_\infty : X_\infty \rightarrow X_\infty$ by

$$(\pi^j A_\infty b)_\alpha = \frac{1}{\alpha \cdot \lambda} (\pi^j b)_\alpha,$$

so that $A_\infty = (A_\infty^\dagger)^{-1}$, hence it makes sense to write A_∞^{-1} instead of $A_\infty^\dagger$. We then define

$$Ab = A_N \pi_N b + A_\infty \pi_\infty b.$$

Having obtained a numerical approximation $\bar{a}$, and having defined the operators $A^\dagger$ and A , we may apply the radii polynomial approach of Theorem 7.6.2 to find $\tilde{a} \in X = (\ell_\nu^1)^n$ such that $F(\tilde{a}) = 0$.

Rather than presenting the general bounds as in Section 8.2, we introduce the bounds in the context of the Lorenz system.

9.5 Applications to the Lorenz system

We begin by applying the radii polynomial approach to compute one-dimensional local stable and unstable manifolds of equilibria in the Lorenz system.

9.5.1 One-dimensional manifolds

Recall from Section 9.2.1 that looking for a local one-dimensional (un)stable manifold attached to an equilibrium $\tilde{x} \in \mathbb{R}^3$ is equivalent to solve the zero finding problem $F(a) = 0$, where each component of $F(a) = (F_1(a), F_2(a), F_3(a))$ is given component-wise by

$$(F_j(a))_k = \begin{cases} (a_j)_0 - \tilde{x}_j, & k = 0, \\ (a_j)_1 - \xi_j, & k = 1, \\ \lambda k(a_j)_k - (\phi_j(a))_k, & k \geq 2, \end{cases}$$

where λ denotes an eigenvalue of $Df(\tilde{x})$ with ξ an associated eigenvector, and where

$$\begin{pmatrix} (\phi_1(a))_k \\ (\phi_2(a))_k \\ (\phi_3(a))_k \end{pmatrix} \stackrel{\text{def}}{=} \begin{pmatrix} \sigma((a_2)_k - (a_1)_k) \\ \rho(a_1)_k - (a_2)_k - (a_1 * a_3)_k \\ -\beta(a_3)_k + (a_1 * a_2)_k \end{pmatrix}, \quad k \geq 2.$$

Here, $m = 1$ (one-dimensional manifold) and $n = 3$ (dimension of the vector field). We fix a Taylor order N , and consider a finite dimensional reduction $F^{(N)} : \mathbb{R}^{3(N+1)} \rightarrow \mathbb{R}^{3(N+1)}$. Using Newton's method, assume we computed an approximate zero $\bar{a}^{(N)} = (\bar{a}_1^{(N)}, \bar{a}_2^{(N)}, \bar{a}_2^{(N)})$ such that $F^{(N)}(\bar{a}^{(N)}) \approx 0$.

Let us now look more concretely at how $A^\dagger$ and A act component-wise. We write $b = (b_1, b_2, b_3)$ and $b^{(N)} = \iota_N \pi_N b = (b_1^{(N)}, b_2^{(N)}, b_3^{(N)})$. We decompose $A^\dagger$ as

$$\pi^j A^\dagger b = A_{j,1}^\dagger b_1 + A_{j,2}^\dagger b_2 + A_{j,3}^\dagger b_3,$$

with

$$(A_{j,j'}^\dagger b_{j'})_k = \begin{cases} \left(D_{j'} F_j^{(N)}(\bar{a}^{(N)}) b_{j'}^{(N)} \right)_k, & 0 \leq k \leq N \\ \delta_{j,j'} \lambda k (b_{j'})_k, & k > N. \end{cases}$$

Here $D_{j'} F_j^{(N)}$ is shorthand for $D_{a_{j'}^{(N)}} F_j^{(N)}$. One may view $A_{j,j'}^\dagger$ as an infinite matrix

$$A_{j,j'}^\dagger \stackrel{\text{def}}{=} \begin{bmatrix} D_{j'} F_j^{(N)}(\bar{a}^{(N)}) & \mathbf{0}^T & \mathbf{0}^T & \mathbf{0}^T \\ \mathbf{0} & \lambda(N+1)\delta_{j,j'} & 0 & 0 \\ \mathbf{0} & 0 & \lambda(N+2)\delta_{j,j'} & 0 \\ \mathbf{0} & 0 & 0 & \ddots \end{bmatrix},$$

where $\mathbf{0} = (0, 0, \dots, 0, 0) \in \mathbb{R}^{N+1}$. The infinite diagonal is thus present only when $i = j$.

Next, we note that the Jacobian matrix $DF^{(N)} = DF^{(N)}(\bar{a}^{(N)})$ can be written as a decomposition in block matrix

$$DF^{(N)} = \begin{pmatrix} DF_{1,1}^{(N)} & DF_{1,2}^{(N)} & DF_{1,3}^{(N)} \\ DF_{2,1}^{(N)} & DF_{2,2}^{(N)} & DF_{2,3}^{(N)} \\ DF_{3,1}^{(N)} & DF_{3,2}^{(N)} & DF_{3,3}^{(N)} \end{pmatrix} \in M_{3(N+1)}(\mathbb{R}).$$

Similarly, $A^{(N)}$ can be written as

$$A^{(N)} = \begin{pmatrix} A_{1,1}^{(N)} & A_{1,2}^{(N)} & A_{1,3}^{(N)} \\ A_{2,1}^{(N)} & A_{2,2}^{(N)} & A_{2,3}^{(N)} \\ A_{3,1}^{(N)} & A_{3,2}^{(N)} & A_{3,3}^{(N)} \end{pmatrix} \in M_{3(N+1)}(\mathbb{R}),$$

where $A^{(N)} \approx (DF^{(N)}(\bar{a}^{(N)}))^{-1}$. In term of components we can write

$$\pi^j Ab = A_{j,1}b_1 + A_{j,2}b_2 + A_{j,3}b_3,$$

where

$$(A_{j,j'}b_{j'})_k = \begin{cases} \left(A_{j,j'}^{(N)}b_{j'}^{(N)}\right)_k & 0 \leq k \leq N, \\ \delta_{j,j'} \frac{1}{\lambda k} (b_{j'})_k & k > N. \end{cases}$$

In particular, the operator A_∞ can be written as

$$\pi^j A_\infty b = \Lambda_N \pi_\infty b_j,$$

Similarly as in Section 8.2, define the component-wise operator $\Lambda_N : X_\infty^{1,\nu} \rightarrow X_\infty^{1,\nu}$ by

$$(\Lambda_N c)_k = \begin{cases} 0 & \text{if } 0 \leq k \leq N, \\ \frac{1}{\lambda k} c_k & \text{if } k \geq N+1. \end{cases} \quad (9.51)$$

The approximate inverse A in this case

$$A = \left[\begin{array}{cc|cc|cc} A_{1,1}^{(N)} & 0 & A_{1,2}^{(N)} & 0 & A_{1,3}^{(N)} & 0 \\ 0 & \Lambda_N & 0 & 0 & 0 & 0 \\ \hline A_{2,1}^{(N)} & 0 & A_{2,2}^{(N)} & 0 & A_{2,3}^{(N)} & 0 \\ 0 & 0 & 0 & \Lambda_N & 0 & 0 \\ \hline A_{3,1}^{(N)} & 0 & A_{3,2}^{(N)} & 0 & A_{3,3}^{(N)} & 0 \\ 0 & 0 & 0 & 0 & 0 & \Lambda_N \end{array} \right].$$

The bound Y_0 . For $j = 1, 2, 3$, set

$$Y_0^{(j)} \stackrel{\text{def}}{=} \sum_{k=0}^N \left| \sum_{j'=1}^3 \left(A_{j,j'}^{(N)} F_{j'}^{(N)}(\bar{a}^{(N)}) \right)_k \right| \nu^k + \sum_{k=N+1}^{2N} \left| \frac{1}{\lambda k} (\phi_j(\bar{a}))_k \right| \nu^k,$$

for $j = 1, 2, 3$, and

$$Y_0 \stackrel{\text{def}}{=} \max \left(Y_0^{(1)}, Y_0^{(2)}, Y_0^{(3)} \right).$$

Then $\|AF(\bar{a})\|_X \leq Y_0$.

The bound Z_0 . Let

$$\begin{aligned} B &\stackrel{\text{def}}{=} I - AA^\dagger \\ &= I - (A_N\pi_N + A_\infty\pi_\infty)(A_N^\dagger\pi_N + A_\infty^\dagger\pi_\infty) \\ &= (\pi_N + \pi_\infty)I - (A_NA_N^\dagger\pi_N + A_\inftyA_\infty^\dagger\pi_\infty) \\ &= (I_{X_N} - A_NA_N^\dagger)\pi_N, \end{aligned}$$

since $A_\infty^\dagger = A_\infty^{-1}$. Here I_{X_N} is the identity on X_N . Using the decomposition notation, we set

$$B_{j,j'}^{(N)} \stackrel{\text{def}}{=} \delta_{j,j'} I_{\mathbb{R}^{N+1}} - \sum_{m=1}^n A_{j,m}^{(N)} D_{j'} F_m^{(N)}(\bar{a}^{(N)}) \in M_{N+1}(\mathbb{R}).$$

These matrices contain all nontrivial behaviour of the infinite dimensional linear operator B :

$$\pi^j B = \sum_{j'=1}^n \iota_N^{-1} B_{j,j'}^{(N)} \iota_N \pi_N \pi^{j'}.$$

Set

$$Z_0^{(j,j')} \stackrel{\text{def}}{=} \max_{0 \leq k' \leq N} \frac{1}{\nu^{k'}} \sum_{0 \leq k \leq N} \left| (B_{j,j'}^{(N)})_{k,k'} \right| \nu^k,$$

and

$$Z_0 \stackrel{\text{def}}{=} \max_{j=1,2,3} \sum_{j'=1}^3 Z_0^{(j,j')}.$$

Then $\|I - AA^\dagger\|_{B(X)} \leq Z_0$.

The bound Z_1 . Recall that we look for a bound $\|A[DF(\bar{a}) - A^\dagger]\|_{B(X)} \leq Z_1$. Given arbitrary $h = (h_1, h_2, h_3) \in X$ with $\|h\|_X \leq 1$, set

$$z \stackrel{\text{def}}{=} [DF(\bar{a}) - A^\dagger]h.$$

From the definitions of DF and $A_\infty^\dagger$, we observe that $(\pi^j z)_k = 0$ for $k = 0, \dots, N$, and for $k > N$,

$$(\pi^j z)_k = (D\phi_j(\bar{a})h)_k = \sum_{j'=1}^3 \left(D_{j'}\phi_j(\bar{a})\pi^{j'}h \right)_k = \sum_{j'=1}^3 (D_{j'}\phi_j(\bar{a})h_{j'})_k,$$

where

$$\sum_{j'=1}^3 D_{j'} \phi_j(\bar{a}) h_{j'} = \begin{cases} \sigma(h_2 - h_1), & j = 1, \\ \rho h_1 - h_2 - \bar{a}_1 * h_3 - h_1 * \bar{a}_3, & j = 2, \\ -\beta h_3 + \bar{a}_1 * h_2 + h_1 * \bar{a}_2, & j = 3. \end{cases}$$

Hence,

$$\begin{aligned} \|\pi^j(Az)\|_{1,\nu} &= \|A_{j,j} \pi_\infty \pi^j z\|_{1,\nu} = \sum_{k \geq N+1} \frac{1}{|\lambda|k} |(\pi^j z)_k| \nu^k \\ &\leq \frac{1}{|\lambda|(N+1)} \sum_{k \geq N+1} |(D\phi_j(\bar{a})h)_k| \nu^k \\ &\leq \frac{1}{|\lambda|(N+1)} \left\| \sum_{j'=1}^3 D_{j'} \phi_j(\bar{a}) \pi^j h \right\|_{1,\nu} \\ &\leq \frac{1}{|\lambda|(N+1)} \begin{cases} |\sigma|(\|h_2\|_{1,\nu} + \|h_1\|_{1,\nu}), & j = 1, \\ |\rho|(\|h_1\|_{1,\nu} + \|h_2\|_{1,\nu} + \|\bar{a}_1 * h_3\|_{1,\nu} + \|h_1 * \bar{a}_3\|_{1,\nu}), & j = 2, \\ |\beta|(\|h_3\|_{1,\nu} + \|\bar{a}_1 * h_2\|_{1,\nu} + \|h_1 * \bar{a}_2\|_{1,\nu}), & j = 3, \end{cases} \\ &\leq \frac{1}{|\lambda|(N+1)} \begin{cases} 2|\sigma|, & j = 1, \\ |\rho| + 1 + \|\bar{a}_1\|_{1,\nu} + \|\bar{a}_3\|_{1,\nu}, & j = 2, \\ |\beta| + \|\bar{a}_1\|_{1,\nu} + \|\bar{a}_2\|_{1,\nu}, & j = 3. \end{cases} \end{aligned}$$

Setting

$$Z_1 = \frac{1}{|\lambda|(N+1)} \max \{2|\sigma|, |\rho| + 1 + \|\bar{a}_1\|_{1,\nu} + \|\bar{a}_3\|_{1,\nu}, |\beta| + \|\bar{a}_1\|_{1,\nu} + \|\bar{a}_2\|_{1,\nu}\},$$

then $\|A[DF(\bar{a}) - A^\dagger]\|_{B(X)} \leq Z_1$.

The bound Z_2 . Recall that we look for the bound $\|A[DF(b) - DF(\bar{a})]\|_{B(X)} \leq Z_2 r$, for all $b \in \overline{B_r(\bar{a})}$. Consider $h \in X$ with $\|h\|_X \leq 1$, and letting $b = \bar{a} + r\hat{h}$, where $\|\hat{h}\|_X \leq 1$, then

$$\pi^j[DF(b) - DF(\bar{a})]h = \begin{cases} 0, & j = 1 \\ \left(-\hat{h}_1 * h_3 - \hat{h}_3 * h_1\right) r, & j = 2 \\ \left(\hat{h}_1 * h_2 + \hat{h}_2 * h_1\right) r, & j = 3 \end{cases}$$

so that for $j = 2, 3$, $\|\pi^j[DF(b) - DF(\bar{a})]h\|_{1,\nu} \leq 2r$. Hence, letting $Z_2 = 2\|A\|_{B(X)}$, $\|A[DF(b) - DF(\bar{a})]\|_{B(X)} \leq Z_2 r$.

9.6 Hartman-Grobman-Theorem

Definition 9.6.1. A vector field $f \in C^1(\mathbb{R}^n, \mathbb{R}^n)$ satisfying $f(0) = 0$ is called *locally topologically conjugate* to its linearization $A \stackrel{\text{def}}{=} Df(0)$ at the origin if there is a homeomorphism $h : U \rightarrow V$ of neighborhoods of the origin such that the flows of f and A are locally conjugated by h , that is

$$h(\varphi(t, x)) = e^{At}h(x),$$

whenever $x \in U$, $t \in \mathbb{R}$, and both sides of the conjugacy equation are defined.

The main goal of this section is to prove the Hartman-Grobman Theorem. While its proof is classical and can be found for instance in [10, 34], we present a proof based on the radii polynomial approach.

The Hartman-Grobman theorem stated below guarantees that the dynamics of a flow near a hyperbolic fixed point is given up to a change of coordinates by the associated linear dynamics. The reader can find classical versions of the proof in [10, 34].

Theorem 9.6.2 (Hartman-Grobman Theorem). *Consider $f \in C^1(\mathbb{R}^n, \mathbb{R}^n)$ such that $f(0) = 0$ and $Df(0)$ hyperbolic. Then, f is locally topologically conjugate to its linearization $Df(0)$ at the origin.*

Proof. A radii polynomial proof... □

Corollary 9.6.3. *Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a C^1 function and let $\tilde{x}$ be an equilibrium solution to $\dot{x} = f(x)$.*

(i) *If $\text{Re}(\lambda) < 0$, for all $\lambda \in \sigma(Df(\tilde{x}))$, then $\tilde{x}$ is asymptotically stable.*

(ii) *If there exists $\lambda_0 \in \sigma(Df(\tilde{x}))$ such that $\text{Re}(\lambda_0) > 0$, then $\tilde{x}$ is unstable.*

Proof. Use the Hartman-Grobman Theorem (Theorem 9.6.2). □

9.7 Exercises